

# Classical Theories on Elliptic Partial Differential Equations

Panhuan Shi

## Abstract

These notes are based on material from Professor Xiaoping Yang's course on elliptic partial differential equations at Nanjing University, Professor Jiakun Liu's summer school lectures at Nanjing University, and my own reading of several standard references. They collect and organize some classical topics in elliptic PDE, including ABP-maximum principle, the moving plane method, regularity theory, De Giorgi–Nash–Moser estimates, and the Krylov–Safonov and Evans–Krylov theories. The emphasis is on the main ideas behind the estimates and on the connections among different regularity methods.

## Contents

|          |                                                           |           |
|----------|-----------------------------------------------------------|-----------|
| <b>1</b> | <b>Gradient Estimate of Semilinear Elliptic Equations</b> | <b>3</b>  |
| 1.1      | Interior Gradient Estimate . . . . .                      | 3         |
| 1.2      | Boundary Gradient Estimate . . . . .                      | 4         |
| <b>2</b> | <b>Alexandroff Maximum Principle</b>                      | <b>7</b>  |
| 2.1      | Contact Set and Normal Mapping . . . . .                  | 7         |
| 2.2      | A Key Lemma . . . . .                                     | 7         |
| 2.3      | Alexandroff Maximum Principle . . . . .                   | 9         |
| 2.4      | Application of Alexandroff Maximum Principle . . . . .    | 12        |
| <b>3</b> | <b>Moving Plane Method</b>                                | <b>15</b> |
| 3.1      | Varadhan Volume Maximum Principle . . . . .               | 15        |
| 3.2      | Moving Plane Method . . . . .                             | 16        |
| <b>4</b> | <b>Morrey Theorem and Campanato Space</b>                 | <b>22</b> |
| 4.1      | Morrey Theorem - Growth of Local Integral . . . . .       | 22        |
| 4.2      | Campanato Space . . . . .                                 | 24        |
| 4.3      | John-Nirenberg Theorem and BMO space . . . . .            | 26        |
| <b>5</b> | <b>Regularity of Solutions</b>                            | <b>27</b> |
| 5.1      | Hölder continuity of solutions . . . . .                  | 27        |
| <b>6</b> | <b><math>L^p</math> Theory for Elliptic Equations</b>     | <b>32</b> |
| 6.1      | Interpolation Theorem . . . . .                           | 32        |
| 6.2      | $L^p$ estimates for Poisson Equation . . . . .            | 34        |
| 6.3      | Calderon-Zygmund Estimates in $L^p$ cases . . . . .       | 37        |
| <b>7</b> | <b>De Giorgi–Nash–Moser Estimates</b>                     | <b>40</b> |
| 7.1      | Local boundedness . . . . .                               | 40        |
| 7.2      | Higher Integrability . . . . .                            | 47        |
| 7.3      | Hölder continuity for homogeneous equations . . . . .     | 48        |
| 7.4      | Weak Harnack and Harnack inequalities . . . . .           | 51        |

|          |                                               |           |
|----------|-----------------------------------------------|-----------|
| <b>8</b> | <b>Krylov–Safonov and Evans–Krylov Theory</b> | <b>55</b> |
| 8.1      | Krylov–Safonov theory . . . . .               | 55        |
| 8.2      | Evans–Krylov theory . . . . .                 | 59        |
|          | <b>References</b>                             | <b>62</b> |

# 1 Gradient Estimate of Semilinear Elliptic Equations

In this note, we consider a domain  $\Omega$  in  $\mathbb{R}^n$  and a strictly elliptic semi-linear operator

$$L_0 u = a^{ij} u_{ij} + b^i u_i = f(x, u).$$

## 1.1 Interior Gradient Estimate

**Theorem 1.1.** *Let  $u$  be a  $C^3(\Omega) \cap C^1(\bar{\Omega})$  solution,  $a^{ij} \in C^1(\bar{\Omega})$ , and  $f \in C^1(\bar{\Omega} \times \mathbb{R})$ . Then*

$$\sup_{\Omega} |Du| \leq \sup_{\partial\Omega} |Du| + C,$$

where  $C = C(\lambda, \Omega, \|u\|_{L^\infty}, f, a^{ij}, b^i)$ .

*Proof.* The proof is a use of classical Bernstein Method.

1. To compute  $L_0(|Du|^2)$  and  $L_0(u^2)$ .

$$L_0(|Du|^2) = 2a^{ij} u_{kj} u_{ki} + 2a^{ij} u_k u_{kj} + 2b_i u_k u_{ki}$$

Take the derivative with respect to  $x_k$  to  $L_0 u = f$ .

$$a^{ij} u_{jk} = (f(x, u))_k + a_k^{ij} u_{ij} - b_k^i u_i - b^i u_{ik}$$

Combining the formula above, we obtain

$$\begin{aligned} L_0(|Du|^2) &= 2a^{ij} u_{kj} u_{ki} \\ &\quad + 2u_k (f_k(x, u) + f_u(x, u) u_k - a_k^{ij} u_{ij} - b_k^i u_i - b^i u_{ik}) \\ &\quad + 2b^i u_k u_{ki} \end{aligned}$$

Note that  $a^{ij} u_{kj} u_{ki} \geq \lambda |D^2 u|^2$ , we have

$$L_0(|Du|^2) \geq 2\lambda |D^2 u|^2 - C |Du|^2 - C$$

(Using Young's inequality to control the term  $2a^{ij} u_k u_{kj} + 2b_i u_k u_{ki}$ ) To compute  $L_0(u^2)$ , we have

$$\begin{aligned} L_0(u^2) &= 2a^{ij} u u_{ij} + 2a^{ij} u_i u_j + 2b^i u u_i \\ &= 2u f(x, u) + 2a^{ij} u_i u_j \geq 2\lambda |Du|^2 - C \end{aligned}$$

As a result, we need to add some terms to  $L_0(|Du|^2 + \alpha u^2 + \text{some terms})$  to apply the maximum principle. On the other hand, assume  $\Omega \subset \{x : x_1 > 0\}$  (by a proper rotation). Then

$$L_0(e^{\beta x_1}) = \beta^2 a^{11} e^{\beta x_1} + 2\beta b^1 e^{\beta x_1} \geq \lambda \beta^2 - C \beta$$

2. We use the auxiliary function  $w(x) = |Du|^2 + \alpha u^2 + e^{\beta x_1}$  with  $\alpha$  and  $\beta$  to be determined positive constants. Through the computation above, we have

$$\begin{aligned} L_0 w &\geq 2\lambda |D^2 u|^2 - C |Du|^2 - C + 2\alpha \lambda |Du|^2 - 2\alpha C + \lambda \beta^2 - C \beta \\ &\geq \lambda |D^2 u|^2 + (2\alpha \lambda - C) |Du|^2 + \lambda \beta^2 - 2\alpha C - C \beta - C > 0 \end{aligned}$$

for sufficiently large  $\alpha$  and  $\beta$ .

By maximum principle, we have

$$\sup_{\Omega} w \leq \sup_{\partial\Omega} w \leq \sup_{\partial\Omega} |Du|^2 + C$$

□

Then we derive a theorem of interior gradient estimate.

**Theorem 1.2** (An Interior Gradient Estimate). *Let  $u$  be a  $C^3(\Omega)$  solution,  $a^{ij} \in C^1(\Omega)$ , and  $f \in C^1(\Omega \times \mathbb{R})$ . Then for any  $\Omega' \subset\subset \Omega$ , we have*

$$\sup_{\Omega'} |Du| \leq C$$

where  $C = C(\lambda, \text{dist}(\Omega', \partial\Omega), \|u\|_{L^\infty}, f, a^{ij}, b^i)$ .

*Proof.* The proof relies on a choice of cut-off function  $\eta \in C_0^\infty(\Omega)$  such that  $\eta = 1$  in  $\Omega'$  and  $\eta = 0$  near  $\partial\Omega$ . Then we apply the previous theorem to  $\eta u$ . Let

$$w(x) = \eta |Du|^2 + \alpha u^2 + e^{\beta x_1}$$

Direct computation shows that

$$\begin{aligned} L_0(\eta |Du|^2) &= \eta L_0(|Du|^2) + |Du|^2 L_0 \eta + 2a^{ij} \eta_i (|Du|^2)_j \\ &\geq \lambda \eta |D^2 u|^2 - C |Du|^2 - C + L_0 \eta |Du|^2 - 2|a^{ij} \eta_i| |Du| |D^2 u| \end{aligned}$$

Note that  $|a^{ij} \eta_i| |Du| |D^2 u| \geq -\varepsilon |D\eta|^2 |D^2 u|^2 - C_\varepsilon |Du|^2$ . We can choose  $\eta$  such that  $|D\eta|^2 \leq C\eta$ . Then we have

$$L_0(\eta |Du|^2) \geq \lambda \eta |D^2 u|^2 - \varepsilon \eta |D^2 u|^2 - C |Du|^2$$

Then the proof follows similarly as the previous theorem. □

## 1.2 Boundary Gradient Estimate

The next theorem is about the Lipschitz continuity of the solution up to the boundary.

**Theorem 1.3** (A Boundary Lipschitz Estimate). *Assume  $\Omega$  is a bounded domain in  $\mathbb{R}^n$  and satisfies the uniform exterior sphere condition, and  $u \in C^2(\Omega) \cap C(\bar{\Omega})$  satisfies the Dirichlet problem*

$$\begin{cases} L_0 u = a^{ij} u_{ij} + b^i u_i = f(x, u) & \text{in } \Omega \\ u = \varphi & \text{on } \partial\Omega \end{cases}$$

where  $\varphi \in C^2(\partial\Omega)$  and  $f(x, u) \in C^1(\Omega \times \mathbb{R})$ . Then there exists a constant  $C$  depending on  $\lambda, \Omega, \|u\|_{L^\infty}, f, a^{ij}, b^i$  and  $\|\varphi\|_{C^2}$  such that

$$|u(x) - u(x_0)| \leq C|x - x_0|, x \in \Omega, x_0 \in \partial\Omega$$

*Proof.* Without loss of generality, we assume  $u|_{\partial\Omega} = 0$  (Extend  $\varphi$  to the whole domain and let  $v = u - \varphi$ ).

To construct  $w \geq 0$  such that

$$\begin{cases} L_0(\pm u - w) \geq 0, & \text{in } \Omega \\ \pm u - w \leq 0, w(x_0) = 0, & \text{on } \partial\Omega \end{cases}$$

Such function is called a barrier function.

Assume such function exists first, then apply maximum principle to  $\pm u - w$  and we obtain  $|u| \leq w$  in  $\Omega$ .

If  $|\nabla w(x)| \leq C$  holds uniformly near  $x_0$ , then

$$|u(x)| \leq w(x) - w(x_0) \leq C|x - x_0|,$$

i.e.  $u$  is Lipschitz continuous up to the boundary.

**Construction of barrier function**  $w(x)$ . The existence of  $w(x)$  relies on the uniform exterior sphere condition, which gives a uniform radius  $R > 0$  for the exterior tangent ball at each boundary point.

Fix  $x_0 \in \partial\Omega$  and choose an exterior ball  $B_R(y)$  with  $\overline{B_R(y)} \cap \bar{\Omega} = \{x_0\}$ . For  $x$  in a neighborhood of  $x_0$  inside  $\Omega$ , set

$$d(x) = \text{dist}(x, \partial B_R(y)) = |x - y| - R.$$

Then  $d(x_0) = 0$ ,  $d(x) > 0$  for  $x \in \Omega$ , and moreover  $|Dd| = 1$  wherever  $d$  is smooth.

We take

$$w(x) = \psi(d(x))$$

for a scalar function  $\psi \in C^2([0, D])$  with  $D = \text{diam}(\Omega)$ . A direct computation gives

$$\begin{aligned} D_i w &= \psi'(d) D_i d, \\ D_{ij} w &= \psi''(d) D_i d D_j d + \psi'(d) D_{ij} d. \end{aligned}$$

Hence

$$L_0 w = \psi''(d) a^{ij} D_i d D_j d + \psi'(d) a^{ij} D_{ij} d + \psi'(d) b^i D_i d.$$

Because  $|Dd| = 1$  and  $a^{ij}$  is uniformly elliptic, we have

$$a^{ij} D_i d D_j d \geq \lambda.$$

On the other hand, using the explicit second derivative of  $d(x) = |x - y| - R$ ,

$$D_{ij} d = \frac{\delta_{ij}}{|x - y|} - \frac{(x_i - y_i)(x_j - y_j)}{|x - y|^3},$$

and writing  $\Lambda = \sup |a^{ij}|$ , one obtains

$$a^{ij} D_{ij} d \leq \frac{n\Lambda - \lambda}{|x - y|} \leq \frac{n\Lambda - \lambda}{R}.$$

Therefore

$$L_0 w \leq \lambda \psi''(d) + \psi'(d) \left( \frac{n\Lambda - \lambda}{R} + \|b\|_{L^\infty} \right).$$

To force  $L_0 w \leq -1$ , choose a positive constant

$$A = \frac{n\Lambda - \lambda}{R} + \|b\|_{L^\infty}$$

and solve

$$\psi'' + A\psi' + 1 = 0.$$

A general solution is

$$\psi(d) = -\frac{1}{A}d + C_1 + C_2 e^{-Ad}.$$

Imposing  $\psi(0) = 0$  gives  $C_1 = -C_2$ , so

$$\psi(d) = -\frac{1}{A}d + \frac{C_2}{A}(1 - e^{-Ad}).$$

Choose  $C_2 \geq e^{AD}$  so that

$$\psi'(d) = C_2 e^{-Ad} - \frac{1}{A} > 0 \quad \text{for } 0 < d \leq D.$$

Then  $\psi''(d) = -AC_2 e^{-Ad} < 0$ , and hence

$$L_0 w \leq -1$$

in a neighborhood of  $x_0$  inside  $\Omega$ .

This  $w$  satisfies  $w(x_0) = 0$ ,  $w > 0$  in  $\Omega$  near  $x_0$ , and has uniformly bounded gradient  $|\nabla w| = \psi'(d) \leq C$ . By the maximum principle we obtain the desired boundary Lipschitz estimate.  $\square$

*Remark 1.4.* The construction is classical.

Associated with the previous two theorems, we have obtained the global gradient estimate.

## 2 Alexandroff Maximum Principle

### 2.1 Contact Set and Normal Mapping

**Definition 2.1.** If  $u \in C(\Omega)$ , where  $\Omega \subset \mathbb{R}^n$  is a bounded domain, the upper contact set of  $u$  is defined as

$$\Gamma_u^+ = \{y \in \Omega : u(x) \leq u(y) + p_y \cdot (x - y), \exists p_y \in \mathbb{R}^n\}.$$

Similarly, the lower contact set of  $u$  is defined as

$$\Gamma_u^- = \{y \in \Omega : u(x) \geq u(y) + p_y \cdot (x - y), \exists p_y \in \mathbb{R}^n\}.$$

**Definition 2.2.** For  $y \in \Gamma_u^+$ , the normal mapping of  $u$  at  $y$  is defined as

$$\chi_u(y) = \{p_y : u(x) \leq u(y) + p_y \cdot (x - y), \forall x \in \Omega\}.$$

If  $u \in C^1(\Omega)$ , then

$$\chi_u(y) = \{Du(y)\}, \quad y \in \Gamma_u^+.$$

1.  $u$  is concave iff  $\Gamma_u^+ = \Omega$
2. If  $u \in C^2(\Omega)$ , then  $D^2u(x) \leq 0, \forall x \in \Gamma_u^+.$

### 2.2 A Key Lemma

Let

$$Lu = a^{ij}D_{ij}u + b^iD_iu + cu \geq f(x) \text{ in } \Omega$$

We need the following two assumptions:

1. 
$$0 < \lambda(x) \leq D^* = (\det(a^{ij}))^{1/n} \leq \Lambda(x)$$

for  $x \in \Omega$ . In fact, if  $(a^{ij})_{n \times n}$  has  $n$  positive eigenvalues:

$$0 < \lambda_1 \leq \dots \leq \lambda_n$$

Then  $D^* = (\lambda_1 \dots \lambda_n)^{1/n}$  and  $\lambda(x) = \lambda_1(x), \Lambda(x) = \lambda_n(x)$ .

- 2.

$$\frac{|b|}{D^*}, \frac{|c|}{D^*}, \frac{|f|}{D^*} \in L^n(\Omega), c(x) \leq 0$$

Then we have the following lemma:

**Lemma 2.3.** Assume  $u \in C^2(\Omega) \cap C(\bar{\Omega})$ , then for  $g \in L_{loc}^1(\mathbb{R}^n), g \geq 0$ , we have

$$\int_{B_{\tilde{M}}(0)} g(p) dp \leq \int_{\Gamma_u^+ \cap \Omega^+} g(Du) |\det(D^2u)| dx$$

where

$$\tilde{M} = \frac{\sup_{\Omega} u - \sup_{\partial\Omega} u^+}{\text{diam}(\Omega)}, \Omega^+ = \{u(x) > 0\}$$

*Proof.* We prove the lemma in two steps.

1. Without loss of generality, we assume  $\sup_{\partial\Omega} u \leq 0$ . Since  $u \in C^1(\Omega)$ , then  $\chi_u(y) = Du(y)$ ,  $y \in \Gamma_u^+$  and  $D^2u(y) \leq 0$ ,  $y \in \Gamma_u^+$ .

Set  $\chi_\varepsilon = \chi_u - \varepsilon Id : \Gamma_u^+ \rightarrow \mathbb{R}^n$ , then

$$D\chi_\varepsilon = D^2u - \varepsilon Id < 0$$

By changing variable, we have

$$\int_{\chi_\varepsilon(\Gamma_u^+ \cap \Omega^+)} g(p) dp = \int_{\Gamma_u^+ \cap \Omega^+} g(\chi_\varepsilon(y)) |\det(D\chi_\varepsilon)| dy$$

Let  $\varepsilon \rightarrow 0$ , we have

$$\lim_{\varepsilon \rightarrow 0} \mathcal{L}^n(\chi_\varepsilon(\Gamma_u^+ \cap \Omega^+)) \geq \mathcal{L}^n(Du(\Gamma_u^+ \cap \Omega^+)).$$

Then it follows that

$$\int_{Du(\Gamma_u^+ \cap \Omega^+)} g(p) dp \leq \int_{\Gamma_u^+ \cap \Omega^+} g(Du) |\det(D^2u)| dx$$

2. To show  $B_{\tilde{M}}(0) \subset Du(\Gamma_u^+ \cap \Omega^+)$ , i.e. for any  $a \in B_{\tilde{M}}(0)$ , there exists  $x \in \Gamma_u^+ \cap \Omega^+$  such that  $Du(x) = a$ .

In fact, assume  $u(0) = m = \sup_{\Omega} u > 0$  for  $|a| < \tilde{M} = m/d$ . Let

$$L(x) = m + a \cdot x$$

, then we have

$$L(0) = m, \quad L(x) \geq m - |a||x| > 0, \quad \forall x \in \Omega.$$

And  $Du(0) = 0$ . As a result, there exists  $x_1$  which is close to the origin such that

$$u(x_1) > L(x_1) > 0.$$

Note that for  $x \in \partial\Omega$ ,

$$u(x) \leq 0 < L(x),$$

so there exists  $\tilde{x} \in \Omega$  such that  $Du(\tilde{x}) = DL(\tilde{x}) = a$ . And it's easy to see that  $\tilde{x} \in \Gamma_u^+ \cap \Omega^+$ . □

*Remark 2.4.* 1. If  $A, B$  are nonnegative definite, then

$$\det A \det B \leq \left( \frac{\text{tr}(AB)}{n} \right)^n.$$

Since  $D^2u \leq 0$  in  $\Gamma_u^+$ , then for any  $x \in \Gamma_u^+$ , we have

$$0 \leq \det(-D^2u) \leq \left( \frac{-a^{ij} D_{ij}u}{nD^*} \right)^n.$$

By lemma 2.3, we have

$$\int_{B_{\tilde{M}}(0)} g(p) dp \leq \int_{\Gamma_u^+ \cap \Omega^+} g(Du) \left( \frac{-a^{ij} D_{ij}u}{nD^*} \right)^n dx.$$

2. Choose  $g = 1$  in lemma 2.3, we have

$$\mathcal{L}^n(B_{\tilde{M}}(0)) \leq \int_{\Gamma_u^+ \cap \Omega^+} \left( \frac{-a^{ij} D_{ij}u}{nD^*} \right)^n dx.$$

which implies

$$\sup_{\Omega} u \leq \sup_{\partial\Omega} u^+ + \text{diam}(\Omega) \omega_n^{-1/n} \left( \int_{\Gamma_u^+ \cap \Omega^+} \left( \frac{-a^{ij} D_{ij}u}{nD^*} \right)^n dx \right)^{1/n}.$$

where  $\omega_n$  is the volume of the unit ball in  $\mathbb{R}^n$ .

### 2.3 Alexandroff Maximum Principle

**Theorem 2.5.** Assume  $\Omega$  is a bounded domain in  $\mathbb{R}^n$  and  $u \in C(\bar{\Omega}) \cap W_{loc}^{2,n}(\Omega)$  such that

$$\forall x \in \Omega, Lu \geq f(x),$$

$L$  is elliptic and (1) holds, then

$$\sup_{\Omega} u \leq \sup_{\partial\Omega} u^+ + C \left\| \frac{f^-}{D^*} \right\|_{L^n(\Gamma_u^+)}$$

where

$$C = C \left( \Omega, n, \left\| \frac{b}{D^*} \right\|_{L^n(\Omega)} \right).$$

*Proof.* We prove the theorem in three steps. We first consider some baby cases (with assumption on the regularity of  $u$  and the ellipticity of  $L$ ), then we make perturbation to deal with the general case.

#### 1. High regularity with $u$ .

We first prove it with the assumption that  $u \in C^2(\Omega) \cap C(\bar{\Omega})$ . Then for  $x \in \Gamma_u^+$ , we have

$$\begin{aligned} \frac{-a^{ij} D_{ij} u}{nD^*} &\leq \frac{b^i D_i u + cu - f}{nD^*} \\ &\leq \frac{b^i D_i u - f^-}{nD^*} \\ &\leq \frac{|b| |Du| + |f^-|}{nD^*} \\ &\leq \frac{1}{nD^*} (Du, \mu) \cdot (|b|, |f^-|/\mu) \\ &\leq \frac{1}{nD^*} (|b|^n + (|f^-|/\mu)^n)^{1/n} \left( |Du|^{\frac{n}{n-1}} + \mu^{\frac{n}{n-1}} \right)^{\frac{n-1}{n}} \end{aligned}$$

where  $\mu > 0$  is some constant to be fixed, and in the last inequality we used Hölder inequality. Then

$$\left( \frac{-a^{ij} D_{ij} u}{nD^*} \right)^n \leq \frac{1}{(nD^*)^n} (|b|^n + (|f^-|/\mu)^n) \left( |Du|^{\frac{n}{n-1}} + \mu^{\frac{n}{n-1}} \right)^{n-1}$$

Let  $g(p) = \left( |p|^{\frac{n}{n-1}} + \mu^{\frac{n}{n-1}} \right)^{1-n}$ , we have

$$\begin{aligned} \int_{B_{\bar{M}}(0)} g(p) dp &\leq \int_{\Gamma_u^+ \cap \Omega^+} g(Du) \left( \frac{-a^{ij} D_{ij} u}{nD^*} \right)^n dx \\ &\leq \int_{\Gamma_u^+ \cap \Omega^+} \frac{|b|^n + (|f^-|/\mu)^n}{(nD^*)^n} dx \\ &\leq n^{-n} \left( \left\| \frac{b}{D^*} \right\|_{\Gamma_u^+ \cap \Omega^+}^n + \left\| \frac{f^-}{D^*} \right\|_{\Gamma_u^+ \cap \Omega^+}^n / \mu^n \right) \end{aligned}$$

Using the following inequality

$$(|p|^{\frac{n}{n-1}} + \mu^{\frac{n}{n-1}})^{n-1} \leq 2^{n-2} (|p|^n + \mu^n)$$

we have

$$g(p) \geq 2^{2-n} (|p|^n + \mu^n)^{-1}$$

Then

$$\begin{aligned}
\int_{B_{\tilde{M}}(0)} g(p) dp &\geq 2^{2-n} \int_{B_{\tilde{M}}(0)} (|p|^n + \mu^n)^{-1} dp \\
&= 2^{2-n} \int_0^{\tilde{M}} \frac{r^{n-1}}{r^n + \mu^n} dr \\
&= 2^{2-n} \omega_n \log \left( \frac{\tilde{M}^n + \mu^n}{\mu^n} \right).
\end{aligned}$$

Then we get the inequality

$$\begin{aligned}
2^{2-n} \omega_n \log \left( \frac{\tilde{M}^n + \mu^n}{\mu^n} \right) &\leq \int_{\Gamma_u^+ \cap \Omega^+} \frac{|b|^n + (|f^-|/\mu)^n}{(nD^*)^n} dx \\
&= n^{-n} \left( \left\| \frac{b}{D^*} \right\|_{L^n(\Gamma_u^+ \cap \Omega^+)}^n + \left\| \frac{f^-}{\mu D^*} \right\|_{L^n(\Gamma_u^+ \cap \Omega^+)}^n \right)
\end{aligned}$$

Then we can solve

$$\tilde{M}^n \leq \mu^n \left( \exp \left( 2^{n-2} n^{-n} \omega_n^{-1} \left( \left\| \frac{b}{D^*} \right\|_{L^n(\Gamma_u^+ \cap \Omega^+)}^n + \left\| \frac{f^-}{\mu D^*} \right\|_{L^n(\Gamma_u^+ \cap \Omega^+)}^n \right) \right) - 1 \right)$$

If  $\left\| \frac{f^-}{D^*} \right\|_{L^n(\Gamma_u^+ \cap \Omega^+)} \neq 0$ , then we may choose  $\mu = \left\| \frac{f^-}{D^*} \right\|_{L^n(\Gamma_u^+ \cap \Omega^+)}$  to get the desired inequality, otherwise we can choose  $\mu$  to be any positive constant and let  $\mu \rightarrow 0$  to get the desired inequality.

## 2. Uniformly elliptic case.

Assume  $L$  is uniformly elliptic, and

$$\sup_{\Omega} \frac{|b|}{\lambda(x)} \leq M < \infty$$

By smooth approximation,  $\forall \varepsilon > 0$ , there exists  $u_m \in C^2(\Omega)$ ,  $\Omega_\varepsilon \subset \subset \Omega$ ,  $|\Omega - \Omega_\varepsilon| < \varepsilon$  such that

$$\begin{aligned}
u_m(x) &\leq \sup_{\Omega} u + \varepsilon, x \in \Omega_\varepsilon \\
\alpha_m &= \|u_m - u\|_{C(\bar{\Omega}_\varepsilon)} \rightarrow 0 \\
\|u_m - u\|_{W^{2,n}(\Omega_\varepsilon)} &\rightarrow 0
\end{aligned}$$

Then by the previous result, we have

$$Lu_m \geq L(u_m - u) + f =: f_m, x \in \Omega$$

Thus

$$\sup_{\Omega_\varepsilon} u_m \leq \sup_{\partial\Omega_\varepsilon} u_m^+ + C \left\| \frac{f_m^-}{D^*} \right\|_{L^n(\Omega_{\varepsilon,m})}$$

where  $\Omega_{\varepsilon,m} = \Gamma_{u_m}^+ \cap \{u_m > 0\}$ . Then

$$\sup_{\Omega_\varepsilon} u_m \leq \sup_{\partial\Omega_\varepsilon} u_m^+ + \varepsilon + C \left\| \frac{(L(u_m - u) + f)^-}{D^*} \right\|_{L^n(\Omega_{\varepsilon,m})}$$

$$\begin{aligned} &\leq \sup_{\partial\Omega_\varepsilon} u_m^+ + \varepsilon + C \left\| \frac{a^{ij} D_{ij}(u_m - u) + b^i D_i(u_m - u)}{D^*} \right\|_{L^n(\Omega_{\varepsilon,m})} + \\ &C \left\| \frac{c}{D^*} \right\|_{L^n(\Omega_{\varepsilon,m})} \alpha_m + C \left\| \frac{f^-}{D^*} \right\|_{L^n(\Omega_{\varepsilon,m})} \end{aligned}$$

Note that since  $\frac{a^{ij}}{D^*} \leq \frac{\Lambda}{\lambda}$ ,

$$\begin{aligned} \left\| \frac{a^{ij} D_{ij}(u_m - u) + b^i D_i(u_m - u)}{D^*} \right\|_{L^n(\Omega_{\varepsilon,m})} &\leq \sup_{\Omega} \frac{\Lambda}{\lambda} \|D_{ij}(u_m - u)\|_{L^n(\Omega_{\varepsilon,m})} \\ &\quad + M \|D_i(u_m - u)\|_{L^n(\Omega_{\varepsilon,m})} \end{aligned}$$

Then we have

$$\begin{aligned} \sup_{\Omega_\varepsilon} u_m &\leq \sup_{\partial\Omega_\varepsilon} u^+ + \varepsilon \\ &\quad + C(\sup_{\Omega} \frac{\Lambda}{\lambda} + M) \|u_m - u\|_{W^{2,n}(\Omega_\varepsilon)} + C \left\| \frac{c}{D^*} \right\|_{L^n(\Omega_{\varepsilon,m})} \alpha_m + C \left\| \frac{f^-}{D^*} \right\|_{L^n(\Omega_{\varepsilon,m})} \end{aligned}$$

Take limit as  $m \rightarrow \infty$ ,

$$\sup_{\Omega_\varepsilon} u \leq \sup_{\partial\Omega_\varepsilon} u^+ + \varepsilon + C \left\| \frac{f^-}{D^*} \right\|_{L^n(\Omega_\varepsilon)}$$

Take limit as  $\varepsilon \rightarrow 0$ , we have

$$\sup_{\Omega} u \leq \sup_{\partial\Omega} u^+ + C \left\| \frac{f^-}{D^*} \right\|_{L^n(\Gamma^+)}$$

### 3. The general case.

To use the previous result, we need to make a perturbation to  $L$  to make it uniformly elliptic. Let

$$L_\eta = L + \eta(\Lambda(x) + |b|)\Delta$$

Then the principal part of  $L_\eta$  is

$$a_\eta^{ij} = a^{ij} + \eta(\Lambda(x) + |b|)\delta_{ij}$$

where the maximum eigenvalue and minimum eigenvalue of  $a_\eta^{ij}$  are  $\Lambda_\eta(x) = \Lambda(x) + \eta(\Lambda(x) + |b|)$  and  $\lambda_\eta(x) = \lambda(x) + \eta(\Lambda(x) + |b|)$  respectively. Then

$$\frac{\Lambda_\eta(x)}{\lambda_\eta(x)} = \frac{\Lambda(x) + \eta(\Lambda(x) + |b|)}{\lambda(x) + \eta(\Lambda(x) + |b|)}$$

Since

$$\lim_{\frac{\Lambda(x)+|b|}{\lambda(x)} \rightarrow \infty} \frac{\Lambda_\eta(x)}{\lambda_\eta(x)} \leq 1 + \frac{1}{\eta}$$

and

$$\lim_{\frac{\Lambda(x)+|b|}{\lambda(x)} \rightarrow \infty} \frac{|b|}{\lambda(x)} \leq \eta$$

Then we can apply the previous result to  $L_\eta$  and  $u$ . (Our purpose is to deal with the case that  $\frac{\Lambda(x)+|b|}{\lambda(x)}$  is not uniformly bounded, thus we need to investigate the limit as  $\frac{\Lambda(x)+|b|}{\lambda(x)} \rightarrow \infty$ ).

We get that

$$L_\eta u = \eta(\Lambda(x) + |b|)\Delta u + Lu \geq \eta(\Lambda(x) + |b|)\Delta u + f =: f_\eta$$

Then

$$\begin{aligned} \sup_{\Omega} u &\leq \sup_{\partial\Omega} u^+ + \tilde{C} \left\| \frac{f_\eta^-}{D_\eta^*} \right\|_{L^n(\Gamma^+)} \\ &\leq \sup_{\partial\Omega} u^+ + \tilde{C} \left\| \frac{f^-}{D_\eta^*} \right\|_{L^n(\Gamma^+)} + \tilde{C} \left\| \frac{\eta(\Lambda(x) + |b|)\Delta u}{D_\eta^*} \right\|_{L^n(\Gamma^+)} \end{aligned}$$

where  $\tilde{C} = \tilde{C}(\Omega, n, \left\| \frac{b}{D_\eta^*} \right\|_{L^n(\Gamma^+)})$

(Note that despite  $\tilde{C}$  relies on  $\eta$  since it relies on  $D_\eta^*$ , however, from the explicit expression of  $C$  in the previous case we can see that we can have a uniform bound for  $\tilde{C}$  if  $\eta$  is sufficiently small, thus we can ignore the dependence of  $\tilde{C}$  on  $\eta$ ).

Let  $\eta \rightarrow 0$ , by dominated convergence theorem, we have

$$\sup_{\Omega} u \leq \sup_{\partial\Omega} u^+ + C \left\| \frac{f^-}{D^*} \right\|_{L^n(\Gamma^+)}$$

□

## 2.4 Application of Alexandroff Maximum Principle

### 2.4.1 Quasilinear Elliptic Equations

We can use Alexandroff maximum principle to investigate quasilinear elliptic equations.

**Theorem 2.6.** *Assume  $u \in C^2(\Omega) \cap C(\bar{\Omega})$  satisfying the quasilinear elliptic equation*

$$Q(u) = a^{ij}(x, u, Du)D_{ij}u + b(x, u, Du) = 0$$

where  $a^{ij} \in C(\Omega, \mathbb{R}, \mathbb{R}^n)$  and is uniformly elliptic.

If there exists non-negative  $g \in L_{loc}^1(\mathbb{R}^n)$ ,  $h \in L^n(\Omega)$  such that

$$\frac{|b(x, \xi, p)|}{nD^*} \leq \frac{h(x)}{g(p)}$$

and

$$\int_{\Omega} h^n(x) dx < \int_{\mathbb{R}} g^n(y) dy =: g_\infty$$

Then, there exists a constant  $C$  depending on  $\Omega, g, h$  such that

$$\sup_{\Omega} |u| \leq \sup_{\partial\Omega} |u| + C \text{diam}(\Omega)$$

*Proof.* Assume  $Q(u) \geq 0$ , then

$$-a^{ij}D_{ij}u \leq b$$

Since  $D_{ij}u \leq 0$  in  $\Gamma_u^+$ , then  $b(x, u, Du) \geq 0$  in  $\Gamma_u^+$ .

By the Key Lemma, we have

$$\int_{B(\tilde{M})} g^n(p) dp \leq \int_{\Gamma_u^+ \cap \Omega^+} g^n(Du) \left( \frac{-a^{ij}D_{ij}u}{nD^*} \right)^n dx$$

$$\begin{aligned}
&\leq \int_{\Gamma_u^+ \cap \Omega^+} g^n(Du) \left( \frac{|b|}{nD^*} \right)^n dx \\
&\leq \int_{\Gamma_u^+ \cap \Omega^+} h^n(x) dx < g_\infty
\end{aligned}$$

which implies that there exists  $C = C(g, h)$  such that  $\tilde{M} \leq C < \infty$ , i.e.

$$\sup_{\Omega} u \leq \sup_{\Omega} u^+ + C \text{diam}(\Omega).$$

Note that  $-u$  satisfies

$$a^{ij}(x, u, Du) D_{ij}(-u) - b(x, u, Du) = 0$$

, then we can similarly show that

$$\sup_{\Omega} (-u) \leq \sup_{\partial\Omega} (-u)^+ + C \text{diam}(\Omega)$$

Thus we have

$$\sup_{\Omega} |u| \leq \sup_{\partial\Omega} |u| + C \text{diam}(\Omega)$$

□

#### 2.4.2 Mean Curvature Equation

Consider  $u \in C^2(\Omega) \cap C(\bar{\Omega})$  is a solution to the prescribed mean curvature equation:

$$\text{div} \frac{Du}{\sqrt{1 + |Du|^2}} = nH(x)$$

or equivalently

$$(1 + |Du|^2) \Delta u - D_i u D_j u D_{ij} u = nH(x)(1 + |Du|^2)^{3/2}.$$

If we let  $H = 0$ , then the equation becomes the minimal surface equation. It is a quasilinear elliptic equation, where

$$a^{ij} = (1 + |p|^2) \delta^{ij} - p^i p^j, b = -nH(x)(1 + |p|^2)^{3/2}$$

Then

$$D^* = (1 + |p|^2)^{\frac{n-1}{n}}$$

and

$$\frac{|b|}{nD^*} = H(x)(1 + |Du|^2)^{\frac{n+2}{2n}}$$

Take

$$h(x) = H(x), g(p) = (1 + |p|^2)^{-\frac{n+2}{2n}}$$

If

$$H_0 := \int_{\Omega} H(x)^n dx < \int_{\mathbb{R}^n} \frac{1}{(1 + |y|^2)^{\frac{n+2}{2}}} dy =: g_\infty$$

then by the previous theorem, we have

$$\sup_{\Omega} |u| \leq \sup_{\partial\Omega} |u| + C(H_0) \text{diam}(\Omega)$$

### 2.4.3 Monge-Ampere Equation

We consider the Monge-Ampere equation

$$\det D^2u = f(x, u, Du), x \in \Omega$$

where  $f \in C(\Omega, \mathbb{R}, \mathbb{R}^n)$ .

**Theorem 2.7.** *Assume  $u \in C^2(\Omega) \cap C(\overline{\Omega})$  is a solution to the Monge-Ampere equation, and there exists non-negative  $h \in L^1(\Omega)$ , and  $0 \leq g(p) \in L^1_{loc}(\mathbb{R}^n)$  such that*

$$|f(x, \xi, p)| \leq \frac{h(x)}{g(p)}$$

and

$$\int_{\Omega} |h(x)| < \int_{\mathbb{R}^n} g(p) dp =: g_{\infty}$$

Then there exists a constant  $C$  depending on  $\Omega, h, g$  such that

$$\sup_{\Omega} |u| \leq \sup_{\partial\Omega} |u| + C \text{diam}(\Omega)$$

*Proof.* The proof is similar to the case of quasilinear elliptic equations. □

*Remark 2.8.* We can also consider the prescribed Gauss curvature equation

$$\det(D^2u) = K(x)(1 + |Du|^2)^{\frac{n+2}{2}}$$

we can acquire a similar maximum principle.

### 3 Moving Plane Method

We recall the Alexandroff Maximum Principle:

**Theorem** (Alexandroff Maximum Principle). *Assume  $\Omega$  is a bounded domain in  $\mathbb{R}^n$  and  $u \in C(\bar{\Omega}) \cap W_{loc}^{2,n}(\Omega)$  such that*

$$\forall x \in \Omega, Lu \geq f(x),$$

*$L$  is elliptic and (1) holds, then*

$$\sup_{\Omega} u \leq \sup_{\partial\Omega} u^+ + C \left\| \frac{f^-}{D^*} \right\|_{L^n(\Omega)}$$

where

$$C = C \left( \Omega, n, \left\| \frac{b}{D^*} \right\|_{L^n(\Omega)} \right).$$

#### 3.1 Varadhan Volume Maximum Principle

Consider

$$Lu = a^{ij} D_{ij} u + b^i D_i u + cu.$$

**Theorem 3.1.** *Assume  $u \in C(\bar{\Omega}) \cap W^{2,n}(\Omega)$ , satisfying  $Lu \geq 0$  in  $\Omega$  and  $u \leq 0$  on  $\partial\Omega$ . Then there is a positive constant  $\delta = \delta(n, d, b, c) > 0$  such that if*

$$\left\| \frac{c^+}{D^*} \right\|_{L^n(\Omega)} < \delta,$$

*then  $u \leq 0$  in  $\Omega$ .*

*Proof.* Since

$$a^{ij} D_{ij} u + b^i D_i u - c^- u \geq -c^+ u,$$

by the Alexandroff Maximum Principle

$$\begin{aligned} \sup_{\Omega} u &\leq C \left\| \frac{(-c^+ u)^-}{D^*} \right\|_{L^n(\Omega)} \\ &= C \left\| \frac{c^+ u^+}{D^*} \right\|_{L^n(\Omega)} \\ &\leq C \sup_{\Omega} u^+ \left\| \frac{c^+}{D^*} \right\|_{L^n(\Omega)}. \end{aligned}$$

Choose  $\delta > 0$  such that  $C\delta < 1/2$ , then

$$\sup_{\Omega} u^+ \leq \frac{1}{2} \sup_{\Omega} u^+$$

we get that  $\sup_{\Omega} u^+ = 0$  in  $\Omega$ . □

*Remark 3.2.* Note that if  $c \in L^\infty(\Omega)$ , then the maximum holds if the domain is small enough.

*Remark 3.3.* We have different maximum principles. It is worth making a comparison.

1. Classical maximum principle.(Strictly elliptic)
2. Serrin maximum principle.(Strictly elliptic, no restrictions on  $c(x)$ )

3. Maximum principle for narrow domain

4. Maximum principle for small volume

**Theorem** (Serrin Maximum Principle). *Assume  $u \in C^2(\Omega) \cap C(\bar{\Omega})$  satisfies  $Lu \geq 0$ . If  $u \leq 0$  in  $\Omega$ , then either  $u < 0$  in  $\Omega$  or  $u = 0$  in  $\Omega$ . (Note that this theorem has no restriction on the sign of  $c(x)$ .)*

**Theorem** (Maximum Principle for Narrow Domain). *Let  $d$  be a positive number and  $e$  be a unit vector such that  $|(y - x) \cdot e| < d$  for any  $x, y \in \Omega$ . Then there exists a  $d_0 > 0$ , depending only on  $\lambda$  and the sup-norm of  $b^i$  and  $c^+$  such that if  $d < d_0$*

1. *there exists a  $w \in C^2(\Omega) \cap C^1(\bar{\Omega})$  satisfying  $w > 0$  in  $\bar{\Omega}$  and  $Lw \leq 0$  in  $\Omega$ .*
2. *If  $u \in C^2(\Omega) \cap C(\bar{\Omega})$  satisfies  $Lu \geq 0$  in  $\Omega$ , then  $\frac{u}{w}$  cannot assume in  $\Omega$  its nonnegative maximum unless  $\frac{u}{w} = \text{const}$ . If, in addition,  $\frac{u}{w}$  assumes its nonnegative maximum at  $x_0 \in \partial\Omega$  and  $\frac{u}{w} \neq \text{const}$ , then for any outward direction  $\nu$  at  $x_0$  to  $\partial\Omega$  there holds*

$$\frac{\partial}{\partial \nu} \left( \frac{u}{w} \right) > 0$$

*if  $\partial\Omega$  has the interior sphere property at  $x_0$ .*

3. *As a result, if  $u \in C^2(\Omega) \cap C(\bar{\Omega})$  and  $Lu \geq 0, u \leq 0$  on  $\partial\Omega$ , then  $u \leq 0$  in  $\Omega$ .*

*Note that this theorem also has no restrictions on  $c$ .*

### 3.2 Moving Plane Method

**Lemma 3.4.** *Assume  $\Omega \subset \mathbb{R}^n$  is a bounded domain and convex in the  $x_1$  direction, symmetric with respect to the plane  $x_1 = 0$ . Then, if  $u \in C^2(\Omega) \cap C(\bar{\Omega})$ , and  $u(x) > 0$ , satisfying*

$$\begin{cases} \Delta u + f(u) = 0, & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

*where  $f$  is Lipschitz, we have  $u$  is symmetric with respect to  $x_1$  and*

$$\frac{\partial u}{\partial x_1} < 0$$

*for any  $x = (x_1, \dots, x_n), x_1 > 0$ .*

#### Idea of the proof

1.  $\forall x = (x_1, x') \in \Omega$ . To show  $\forall x_1 > 0, x_1^* < x_1$ , with  $x_1^* + x_1 > 0, u(x_1, x') < u(x_1^*, x')$ , which can prove the partial derivation about  $x_1$  is non-positive. It is actually negative by the Hopf Lemma.
2. Let  $x_1^* \rightarrow -x_1$ , we can obtain

$$u(x_1, x') \leq u(-x_1, x')$$

Then change the direction and through similar argument one can show that

$$u(-x_1, x') \leq u(x_1, x')$$

since the domain  $\Omega$  is symmetric and the equation is invariant under the change of direction. Then we get the desired result.

*Proof.* 1. Let  $a = \sup_{x \in \Omega} x_1$ . Then for  $0 < \lambda < a$ , define

$$T_\lambda = \{x_1 = \lambda\}, \Sigma_\lambda = \{x \in \Omega : x_1 > \lambda\}$$

And  $\Sigma'_\lambda$  is the reflection domain of  $T_\lambda$ .

By the convexity and symmetry of the domain, we conclude that  $\Sigma'_\lambda \subset \Omega$ . We denote that  $x_\lambda = (2\lambda - x_1, x')$  (the reflectional point).

Let  $w_\lambda(x) = u(x) - u(x_\lambda), \forall x \in \Sigma_\lambda$ . We want to prove  $w_\lambda$  satisfies a certain equation.

From the equation we have

$$\begin{cases} \Delta u + f(u) = 0 \\ \Delta u(x_\lambda) + f(u(x_\lambda)) = 0 \end{cases}$$

then  $\Delta w_\lambda = f(u(x_\lambda)) - f(u) = -C(x, \lambda)w_\lambda$  since  $f$  is Lipschitz. i.e.

$$\begin{cases} \Delta w_\lambda + C(x, \lambda)w_\lambda = 0, \forall x \in \Sigma_\lambda \\ w_\lambda(x)|_{\partial\Sigma_\lambda} \leq 0, w_\lambda(x)|_{\partial\Sigma_\lambda} \not\equiv 0 \end{cases}$$

The second inequality can be verified easily by separating  $\partial\Sigma_\lambda$  into two portions.

2. To show  $\forall x \in \Sigma_\lambda, 0 < \lambda < a, w_\lambda(x) < 0$ .

- First we choose  $\lambda$  close to  $a$ , Using the maximum principle for narrow domain,  $w_\lambda(x)|_{\Sigma_\lambda} < 0$ .

Assume  $(\lambda_0, a)$  is the maximal interval such that

$$w_\lambda(x) < 0, \forall x \in \Sigma_\lambda, \lambda_0 < \lambda < a$$

- If  $\lambda_0 > 0$ , by the continuity of  $u$ ,  $\lim_{\lambda \rightarrow \lambda_0} w_\lambda(x) = w_{\lambda_0}(x), \forall x \in \Sigma_{\lambda_0}$ , and  $w_{\lambda_0}(x)|_{\partial\Sigma_{\lambda_0}} \not\equiv 0$

Furthermore, by the Serrin Maximum Principle, we have  $w_{\lambda_0}(x) < 0$  in  $\Sigma_{\lambda_0}$ .

- To show there exists  $\varepsilon > 0$  small enough,

$$w_{\lambda_0 - \varepsilon}|_{\Sigma_{\lambda_0 - \varepsilon}} < 0$$

then there is a contradiction.

In fact, fix  $\delta > 0$ , there exists a compact  $K \subset \Sigma_{\lambda_0}$  and  $\eta > 0$ , such that  $w_{\lambda_0}(x)|_K \leq -\eta < 0$ , and

$$\mathcal{L}^n(\Sigma_{\lambda_0} - K) < \frac{\delta}{2}$$

Then there exists  $\varepsilon > 0$  (continuity), such that

$$w_{\lambda_0 - \varepsilon}(x)|_K \leq -\frac{\eta}{2} < 0, \mathcal{L}^n(\Sigma_{\lambda_0 - \varepsilon} - K) < \delta$$

Then we apply the Varadhan Volume Maximum Principle to the rest of the domain (with volume  $< \delta$ ) to get a contradiction.

□

The following theorem is a direct corollary.

**Theorem 3.5.** Assume  $u \in C(\bar{B}) \cap C^2(B)$  and  $u(x) > 0$  in  $B$  satisfying

$$\begin{cases} \Delta u + f(u) = 0 \text{ in } B \\ u = 0 \text{ on } \partial B \end{cases}$$

where  $f$  is Lipschitz Then  $u$  is radially symmetric and  $\frac{\partial u}{\partial r} < 0$

### 3.2.1 Moving planes at infinity

**Theorem 3.6** (Caffarelli–Gidas–Spruck). *Let  $u$  be a positive solution of*

$$\Delta u + u^{\frac{n+2}{n-2}} = 0 \quad \text{in } \mathbb{R}^n, \quad n \geq 3.$$

*Then  $u$  is radially symmetric with respect to some  $x_0 \in \mathbb{R}^n$ , and*

$$u(x) = \frac{(n(n-2)\lambda^2)^{\frac{n-2}{4}}}{(\lambda^2 + |x - x_0|^2)^{\frac{n-2}{2}}}$$

*for some  $\lambda > 0$ .*

*Remark 3.7.* In 1979, Gidas, Ni and Nirenberg proved in *Symmetry and related properties via the maximum principle* that positive solutions are radially symmetric under suitable decay assumptions at infinity  $u(x) = O(|x|^{2-n})$ . In 1989, Caffarelli, Gidas and Spruck removed this decay assumption in *Asymptotic symmetry and local behavior of semilinear elliptic equations with critical Sobolev growth*.

I suspect that the condition in the problem was intended to be

$$\lim_{|x| \rightarrow \infty} |x|^{n-2} u(x) = C < +\infty,$$

In what follows, I will present the proof of the more general result.

**Proof. Step I: Kelvin Transform of  $u$ .**

Define the Kelvin transform of  $u$  as

$$v(x) = |x|^{2-n} u\left(\frac{x}{|x|^2}\right).$$

The origin is a possible singularity of  $v$  and the singularity is removable if  $u(x) = O(|x|^{2-n})$  at infinity. Moreover, we have  $v(x) = O(|x|^{2-n})$  at infinity.

Direct computation shows that

$$\begin{aligned} \Delta v &= \Delta \left( |x|^{2-n} u\left(\frac{x}{|x|^2}\right) \right) \\ &= \Delta(|x|^{2-n}) u\left(\frac{x}{|x|^2}\right) + 2\nabla(|x|^{2-n}) \cdot \nabla \left( u\left(\frac{x}{|x|^2}\right) \right) + |x|^{2-n} \Delta u\left(\frac{x}{|x|^2}\right) \\ &= (4-2n)|x|^{-n} x_i u_j \left(\frac{x_j}{|x|^2}\right)_i + |x|^{2-n} u_{jk} \left(\frac{x_j}{|x|^2}\right)_i \left(\frac{x_k}{|x|^2}\right)_k + |x|^{2-n} u_j \left(\frac{x_j}{|x|^2}\right)_{ii} \\ &= -\frac{x_i u_i}{|x|^{2+n}} - |x|^{-4-n} \left( u\left(\frac{x}{|x|^2}\right) \right)^{\frac{n+2}{n-2}} + \frac{x_i u_i}{|x|^{2+n}} \\ &= -v^{\frac{n+2}{n-2}} \end{aligned}$$

away from the origin.

We can also prove that  $\Delta v + v^{\frac{n+2}{n-2}} = 0$  in  $\mathbb{R}^n$  in the distribution sense.

We first show that  $v \in L^p(B_1)$ . Fix  $k > \max_{|x|=1} v$ . Let  $\psi(x) \geq 0$  be a nonincreasing smooth function such that

$$\psi(t) = \begin{cases} 1, & t < k \\ 0, & t \geq 2k \end{cases},$$

and set  $\Phi(t) = \int_0^t \psi(s) ds$ . Define the cut-off function  $\eta \equiv 1$  in  $B_1 - B_{2\varepsilon}$  and  $\eta \equiv 0$  in  $B_\varepsilon$ . Then

$$\int_{B_1} \nabla v \cdot \nabla(\eta\psi) = \int_{B_1} \eta\psi v^p + \int_{\partial B_1} u_\nu,$$

where  $\psi = \psi(v)$ , moreover,

$$\int_{B_1} \nabla v \cdot \nabla(\eta\psi) = \int_{B_1} \eta\psi' |\nabla v|^2 + \int_{B_1} \nabla\Phi \cdot \nabla\eta \leq - \int_{B_1} \Phi(v) \Delta\eta = O(\varepsilon^{n-2}).$$

Let  $\varepsilon \rightarrow 0$  we obtain

$$\int_{B_1 \cap \{u < k\}} u^p \leq - \int_{\partial B_1} u_\nu.$$

Letting  $k \rightarrow \infty$ , we conclude that  $v \in L^p(B_1)$ .

Then it suffices to show that

$$\int_{B_1} v \Delta\zeta + \zeta v^p = 0, \quad \zeta \in C_0^\infty(B_1).$$

Using  $\eta\zeta$  as a test function where

Then

$$\int_{B_1} \nabla v \cdot \nabla(\eta\zeta) = \int_{B_1} \eta\zeta v^p,$$

and

$$\int_{B_1} \nabla v \cdot \nabla(\eta\zeta) = - \int_{B_1} v \Delta(\eta\zeta) = - \int_{B_1} v(\zeta \Delta\eta + 2\nabla\eta \cdot \nabla\zeta + \eta \Delta\zeta),$$

hence

$$\left| \int_{B_1} v(\zeta \Delta\eta + 2\nabla\eta \cdot \nabla\zeta) \right| \leq \frac{C}{\varepsilon^2} \int_{B_{2\varepsilon} - B_\varepsilon} u \leq \frac{C}{\varepsilon^2} \cdot \varepsilon^{n(1-\frac{1}{p})} \|v\|_{L^p(B_{2\varepsilon})} \rightarrow 0$$

as  $\varepsilon \rightarrow 0$ .

Therefore  $v$  is a distribution solution and we can assume that  $v$  is lower-semicontinuous.

**Step II: Prepare to move the plane.**

For  $\lambda \in \mathbb{R}$ , we define

$$\Sigma_\lambda = \{x_n > \lambda\}, \quad T_\lambda = \partial\Sigma_\lambda, \quad x_\lambda = (x_1, \dots, 2\lambda - x_n).$$

Then we prove that for  $\lambda$  large enough,

$$w_\lambda(x) := v(x) - v(x_\lambda) \leq 0, \quad x \in \Sigma_\lambda.$$

We notice that  $w_\lambda$  satisfies the equation

$$\Delta w_\lambda(x) + c(x, \lambda) w_\lambda(x) = 0$$

where  $c(x, \lambda) = \frac{v^p - v_\lambda^p}{v - v_\lambda} = p\xi_\lambda^{p-1}$ .

$x_0 = (0, 2\lambda)$  is a possible singularity of  $w_\lambda$ . We show that  $w_\lambda < 0$  if  $\lambda$  is large enough.

Since  $\Delta v = -v^{\frac{n+2}{n-2}} \leq 0$ ,  $v$  is a superharmonic function in the distribution sense, then

$$v(x) \geq \min_{\partial B_1} v =: \varepsilon_0 > 0, \quad x \in B_1$$

We fix a  $\lambda_0$  large enough and  $v(x) \leq \varepsilon_0$  when  $\lambda > \lambda_0$ . It follows that

$$w_\lambda \leq 0, \quad x \in B_1(x_0)$$

For the domain  $\Sigma'_\lambda = \Sigma_\lambda - B_1(x_0)$ , we will show that  $w_\lambda^+ \equiv 0$ .

We choose a cut-off function  $\zeta \in C_0^\infty(B_{2R})$  such that  $\zeta \equiv 1$  on  $B_R$  and  $|\nabla\zeta| \leq \frac{2}{R}$ .

Using  $\phi = \zeta^2 w_\lambda^+$  as the test function, integrating by parts and Young's inequality gives that

$$\int_{\Sigma'_\lambda} \zeta^2 |\nabla w_\lambda^+|^2 \leq \int_{\Sigma'_\lambda} \zeta^2 (w_\lambda^+)^2 c(x, \lambda) + \int_{\Sigma'_\lambda} |\nabla\zeta|^2 (w_\lambda^+)^2$$

or

$$\int_{\Sigma'_\lambda} |\nabla(\zeta w_\lambda^+)|^2 \leq \int_{\Sigma'_\lambda} \zeta^2 (w_\lambda^+)^2 c(x, \lambda) + \int_{\Sigma'_\lambda} |\nabla \zeta|^2 (w_\lambda^+)^2.$$

Using the Sobolev inequality, we get

$$\left( \int_{\Sigma'_\lambda} (\zeta w_\lambda^+)^{\frac{2n}{n-2}} \right)^{\frac{n-2}{n}} \leq \int_{\Sigma'_\lambda} \zeta^2 (w_\lambda^+)^2 c(x, \lambda) + \int_{\Sigma'_\lambda} |\nabla \zeta|^2 (w_\lambda^+)^2.$$

Since  $w_\lambda(x) = O(|x|^{2-n})$  at infinity, we estimate the second term on the right hand side,

$$\int_{\Sigma'_\lambda} |\nabla \zeta|^2 (w_\lambda^+)^2 \leq \frac{C}{R^2} \int_{B_{2R}-B_R} |x|^{4-2n} dx = \frac{C}{R^2} \int_R^{2R} r^{3-n} dr \leq \frac{C}{R^{n-2}} \rightarrow 0$$

as  $R \rightarrow \infty$ . Hence we have

$$\left( \int_{\Sigma'_\lambda} (w_\lambda^+)^{\frac{2n}{n-2}} \right)^{\frac{n-2}{n}} \leq \int_{\Sigma'_\lambda} (w_\lambda^+)^2 c(x, \lambda) \leq \left( \int_{\Sigma'_\lambda} (w_\lambda^+)^{\frac{2n}{n-2}} \right)^{\frac{n-2}{n}} \left( \int_{\Sigma_\lambda^+ \cap \{w_\lambda \geq 0\}} c(x, \lambda)^{\frac{n}{2}} \right)^{\frac{2}{n}}.$$

The last inequality holds by the Hölder inequality. We can easily verify that  $w_\lambda^+ \in L^{\frac{2n}{n-2}}(\Sigma_\lambda^+)$  and  $c(x, \lambda) \in L^{\frac{n}{2}}(\Sigma'_\lambda \cap \{w_\lambda \geq 0\})$ .

It suffices to choose  $\lambda$  large enough ( $\lambda > \lambda_1$ ) such that

$$\int_{\Sigma'_\lambda \cap \{w_\lambda \geq 0\}} c(x, \lambda)^{\frac{n}{2}} \leq C \int_{\Sigma'_\lambda} (v^{p-1})^{\frac{n}{2}} \leq C \int_{\Sigma'_\lambda} |x|^{-2n} dx < 1,$$

where the constant  $C$  relies only on  $n$  and the original solution  $u$ , and we can deduce that  $w_\lambda^+ \equiv 0$  on  $\Sigma'_\lambda$ .

**Step III: The limiting position.**

We define  $\lambda_0$  as

$$\lambda_0 = \inf_{\lambda} \{ \lambda : w_\mu(x) \leq 0, x \in \Sigma_\lambda, \mu \geq \lambda \} =: \inf_{\lambda} \mathcal{S}.$$

Obviously  $\lambda_0 \neq -\infty$  since  $v(x) = O(|x|^{2-n})$ .

Assume that  $w_{\lambda_0} \equiv 0$  doesn't hold in  $\Sigma_{\lambda_0}$ , then by Strong Maximum Principle,  $w_{\lambda_0} < 0$  in  $\Sigma_{\lambda_0}$ .

We consider three subcases.

**Case (i):**  $\lambda_0 > 0$ .

From a similar argument in **Step II** we discover that  $w_\lambda \leq 0$  in the domain  $\Omega$  holds when the integral

$$\int_{\Sigma'_\lambda \cap \Omega} |x|^{-2n} dx$$

is small enough.

Then we choose a suitable small  $\varepsilon_1 > 0$  and large  $R > 0$  such that in the region  $\Omega = \{|x_n - \lambda_0| < \varepsilon\} \cup \{|x| > R\}$  excluding the singularity, there holds

$$\int_{\Sigma'_{\lambda_0-\varepsilon} \cap \Omega} |x|^{-2n} dx$$

is sufficiently small,  $\forall \varepsilon \leq \varepsilon_1$ , i.e.  $w_{\lambda-\varepsilon} \leq 0$  in  $\Omega$ .

For the rest of the region  $\Sigma_{\lambda_0} - \Omega$  (containing the possible singularity  $(0, 2\lambda)$ ), which is a compact set, we can find  $\mu$  such that  $w_{\lambda_0} < -\mu$  in  $\Sigma_{\lambda_0} - \Omega$  since  $w_\lambda$  is upper-semicontinuous with respect to  $x$  and  $\lambda$ , moreover

$$w_{\lambda_0-\varepsilon} < -\frac{\mu}{2}$$

for  $\varepsilon < \varepsilon_1$ , which implies that  $\lambda_0 - \varepsilon \in \mathcal{S}$ , which is a contradiction. Hence we must have  $w_{\lambda_0} \equiv 0$  in  $\Sigma_\lambda$ .

**Case (ii):**  $\lambda_0 < 0$ .

We consider to move the plane from the other direction of  $x_n$ -axis (negative infinity).

If the limiting position  $\lambda'_0 < 0$ , then a similar argument in **Case (i)** gives the symmetry and monotonicity. Otherwise, if  $\lambda'_0 \geq 0$ , we obtain an inequality of the other direction, which derives a contradiction.

**Case (iii):**  $\lambda_0 = 0$ .

We also consider to move the plane from the other direction. Suppose that the limiting position  $\lambda'_0 \neq 0$ , then a similar argument as in **Case (i)(ii)** gives the symmetry with respect to  $x_n = \lambda'_0$  and the monotonicity. If the limiting position  $\lambda'_0 = 0$ , which gives the other direction of the inequality, we also obtain the symmetry with respect to  $x_n = 0$  and the monotonicity.

**Step IV: The radial symmetry.** Since the direction  $x_n$  can be chosen arbitrarily. We conclude that  $v$  is radially symmetric with respect to some point  $x_0$ .

If  $x_0$  is the origin, then  $u$  is radially symmetric as well. If  $x_0$  is not the origin, then the radial symmetry tells that 0 is a removable singularity of  $v$  otherwise  $v$  will have more than one singularity. It implies that  $u(x) = O(|x|^{2-n})$  at infinity. In this situation, just repeat the previous steps (no singularity version) and we can obtain the radial symmetry of  $u$ .

**Step V: The explicit solution.** We assume that

$$u(x) = f(|x - x_0|)$$

for the symmetric center  $x_0$ . Then the PDE is restricted to the ODE

$$f'' + \frac{n-1}{r}f' + f^{\frac{n+2}{n-2}} = 0.$$

It is easy to verify that

$$u(x) = \frac{(n(n-2)\lambda^2)^{\frac{n-2}{4}}}{(\lambda^2 + r^2)^{\frac{n-2}{2}}}$$

is a solution thus is the unique solution by the uniqueness theorem of ODE.

□

## 4 Morrey Theorem and Campanato Space

### 4.1 Morrey Theorem - Growth of Local Integral

Let  $\Omega \subset \mathbb{R}^n$  be a bounded connected domain and  $u \in L^1(\Omega)$ ,  $B_1(x_0) \subset \Omega$ , we can define the mean value of  $u$  over  $B_1(x_0)$  as

$$u_{x_0,r} = \frac{1}{|B_r(x_0)|} \int_{B_r(x_0)} u(y) dy$$

Then we have the following theorem:

**Theorem 4.1** (Morrey Theorem). *Assume  $u \in L^2(\Omega)$ , then there exists  $M > 0, 0 < \alpha \leq 1$ , for any  $B_r(x_0) \subset \Omega$ , we have*

$$\int_{B_r(x_0)} |u(x) - u_{x_0,r}|^2 dx \leq M^2 r^{n+2\alpha}$$

Then  $u \in C^\alpha(\Omega)$ , and for  $\Omega' \subset \subset \Omega$ , we have

$$|u|_{0,\alpha;\Omega'} \leq C(M + \|u\|_{L^2(\Omega)})$$

where  $C = C(n, \alpha, \Omega, \Omega')$

*Proof.* 1. To show  $u_{x,r}$  is convergent as  $r \rightarrow 0, x \in \Omega$ .

In fact, let  $R_0 = \text{dist}(\Omega', \partial\Omega)$ , then for any  $x_0 \in \Omega', 0 < r_1 < r_2 \leq R_0$ , we have

$$|u_{x_0,r_1} - u_{x_0,r_2}|^2 \leq 2(|u(x) - u_{x_0,r_1}|^2 + |u(x) - u_{x_0,r_2}|^2)$$

Integrating over  $B_{r_1}(x_0)$  on both sides, we obtain

$$\begin{aligned} |u_{x_0,r_1} - u_{x_0,r_2}|^2 &\leq \frac{2}{|B_{r_1}(x_0)|} \left( \int_{B_{r_1}(x_0)} |u - u_{x_0,r_1}|^2 dx + \int_{B_{r_1}(x_0)} |u - u_{x_0,r_2}|^2 dx \right) \\ &\leq \frac{2}{|B_{r_1}(x_0)|} \left( \int_{B_{r_1}(x_0)} |u - u_{x_0,r_1}|^2 dx + \int_{B_{r_2}(x_0)} |u - u_{x_0,r_2}|^2 dx \right) \\ &\leq \frac{2M^2}{\omega_n r_1^n} (r_1^{n+2\alpha} + r_2^{n+2\alpha}) \\ &\leq C(n) \left( \frac{r_2}{r_1} \right)^n r_2^{2\alpha} = C(n) (r_2/r_1)^n r_2^{2\alpha}. \end{aligned}$$

Then for  $R \leq R_0$ , choose

$$r_1 = \frac{R}{2^{i+1}}, \quad r_2 = \frac{R}{2^i}.$$

Then

$$|u_{x_0,r_1} - u_{x_0,r_2}| \leq C(n) M 2^{-i\alpha} R^\alpha.$$

For  $h < k$ , summing the estimates for  $i = 0, \dots, k-h-1$  yields

$$\begin{aligned} |u_{x_0,r_h} - u_{x_0,r_k}| &\leq \sum_{i=0}^{k-h-1} \left| u_{x_0,2^{-h+i+1}R} - u_{x_0,2^{-h+i}R} \right| \\ &\leq C(n) M R^\alpha \sum_{i=0}^{k-h-1} 2^{-i\alpha} \leq \frac{c(n,\alpha)}{2^{h\alpha}} M R^\alpha \end{aligned}$$

which indicates that  $u_{x_0,r}$  is convergent as  $r \rightarrow 0, x \in \Omega$ . We denote the limit by  $\hat{u}(x_0)$ .

We note that the limit function is independent of the choice of  $R$ . Actually, we can choose  $r_1 = 2^{-iR}, r_2 = 2^{-i\bar{R}}, 0 < R < \bar{R} < R_0$ , then

$$|u_{x_0, r_1} - u_{x_0, r_2}|^2 \leq C(n)M^2 \left(\frac{\bar{R}}{R}\right)^n 2^{-2i\bar{R}\alpha} \rightarrow 0$$

as  $i \rightarrow \infty$

We have the estimate

$$\begin{aligned} |\hat{u}(x_0) - u_{x_0, r}|^2 &\leq \frac{1}{|B_r(x_0)|} \int_{B_r(x_0)} |u - \hat{u}|^2 dx \\ &\leq C(n, \alpha)Mr^\alpha \end{aligned} \quad (1)$$

2. To show the boundedness and continuity of  $u$ .

By Lebesgue Theorem, we have  $u = \hat{u}$  a.e. in  $\Omega$ . By the absolute continuity of integral, we have  $u_{x, r}$  is continuous with  $x$  for each fixed  $r$ . Then we have that  $u$  is continuous a.e. since  $u(x, r)$  converges to  $\hat{u}(x_0)$  as  $r \rightarrow 0$  uniformly in  $x$  by equation (1).

Moreover, we have that

$$|u(x)| \leq C(n, \alpha)MR^\alpha + |u_{x, R}|$$

which implies

$$\sup_{\Omega'} |u| \leq C(MR_0^\alpha + \|u\|_{L^2(\Omega)})$$

hence  $u$  is bounded in  $\Omega'$ .

3. Finally we prove that  $u$  is Hölder continuous.

For any  $x, y \in \Omega', |x - y| = R \leq \frac{R_0}{2}$ , we have

$$\begin{aligned} |u(x) - u(y)| &\leq |u(x) - u_{x, 2R}| + |u(y) - u_{y, 2R}| + |u_{x, 2R} - u_{y, 2R}| \\ &\leq C(n, \alpha)MR^\alpha + |u_{x, 2R} - u_{y, 2R}| \end{aligned}$$

where

$$|u_{x, 2R} - u_{y, 2R}|^2 \leq 2(|u_{x, 2R} - u(\zeta)|^2 + |u_{y, 2R} - u(\zeta)|^2)$$

Integral over  $B_R(x)$  on both sides, we obtain

$$\begin{aligned} |u_{x, 2R} - u_{y, 2R}|^2 &\leq \frac{2}{|B_R(x)|} \left( \int_{B_{2R}(x)} |u - u_{x, 2R}|^2 + \int_{B_{2R}(y)} |u - u_{y, 2R}|^2 \right) \\ &\leq C(n, \alpha)M^2R^{2\alpha} \end{aligned}$$

Therefore we have

$$|u(x) - u(y)| \leq C(n, \alpha)M|x - y|^\alpha$$

For  $|x - y| \geq \frac{R_0}{2}$ , the estimate is trivial through the boundedness of  $u$ . □

*Remark 4.2.* Note that here the index 2 is not necessary, it can be replaced by any  $p \geq 1$  and correspondingly the index  $n + 2\alpha$  needs to be replaced by  $n + p\alpha$ .

Then we can derive the following result due to Morrey Theorem.

**Corollary 4.3.** Assume  $u \in H_{loc}^1(\Omega)$  satisfying for any  $B_r(x) \subset\subset \Omega, r > 0$ , there exists  $M > 0, 0 < \alpha < 1$  such that

$$\int_{B_r(x)} |Du|^2 \leq M^2 r^{n-2+2\alpha}$$

Then  $u \in C^\alpha(\Omega)$  and for  $\Omega' \subset\subset \Omega$ , we have

$$|u|_{0,\alpha;\Omega'} \leq C(M + \|u\|_{L^2(\Omega)})$$

where  $C = C(n, \alpha, \Omega, \Omega') > 0$ .

*Proof.* The theorem is a easy application of Poincaré inequality.  $\square$

**Remark 4.4.** For  $u \in W_{loc}^{1,p}(\Omega)$ , the index  $n - 2 + 2\alpha$  needs to be replaced by  $n - p + p\alpha$ .

## 4.2 Companato Space

**Definition 4.5** (Domain of Type A). Let  $\Omega \subset \mathbb{R}^n$  be a bounded domain. If there exists  $A > 0$  such that for any  $x \in \Omega, 0 < \rho < \text{dist}(\Omega)$ ,

$$|\Omega \cap B_\rho(x)| \geq A\rho^n$$

then  $\Omega$  is called the type domain of  $A$ .

If you are familiar with the *Real and Complex Analysis* by Walter Rudin, the Type  $A$  domain coincides with the "nicely shrinking" set in Chapter 7 of that book.

**Definition 4.6** (Companato Space). Let  $p \geq 1, \mu \geq 0, u \in L^p(\Omega)$ , then we define

$$[u]_{p,\mu;\Omega} = [u]_{p,\mu} := \sup_{x \in \Omega, 0 < \rho \leq \text{diam}(\Omega)} \left( \rho^{-\mu} \int_{\Omega \cap B_\rho(x)} |u - u_{x,\rho}|^p \right)^{1/p} < +\infty$$

furthermore, we have the norm

$$\|u\|_{\mathcal{L}^{p,\mu}(\Omega)} := [u]_{p,\mu;\Omega} + \|u\|_{L^p(\Omega)}$$

and the space  $\mathcal{L}^{p,\mu}(\Omega)$  is the function space with finite norm, which is called Companato space.

The Companato space is a Banach space.

**Remark 4.7.** 1.  $[u]_{p,\mu;\Omega}$  is a seminorm.

2. If  $\Omega$  is a type domain of  $A$ , then

$$\mathcal{L}^{p,\mu}(\Omega) = \begin{cases} C^\alpha(\bar{\Omega}), n < \mu \leq n + p, \alpha = \frac{\mu-n}{p} \\ \text{const}, \mu > n + p \end{cases}$$

3. If  $p \leq q, \frac{n-\mu}{p} \geq \frac{n-\nu}{q}$ , then

$$\mathcal{L}^{q,\nu}(\Omega) \subset \mathcal{L}^{p,\mu}(\Omega)$$

We also give the definition of the Morrey space.

**Definition 4.8** (Morrey Space). Let  $p \geq 1, \mu \geq 0, u \in L^p(\Omega)$  satisfying

$$\sup_{x \in \Omega, 0 < \rho \leq \text{diam}(\Omega)} \left( \rho^{-\mu} \int_{\Omega \cap B_\rho(x)} |u|^p \right)^{1/p} < +\infty$$

We can define the norm as

$$\|u\|_{L^{p,\mu}(\Omega)} := \sup_{x \in \Omega, 0 < \rho \leq \text{diam}(\Omega)} \left( \rho^{-\mu} \int_{\Omega \cap B_\rho(x)} |u|^p \right)^{1/p}$$

which also defines a Banach space. We called the space  $L^{p,\mu}(\Omega)$  the Morrey space.

**Proposition 4.9.** *We list some properties of the Morrey space:*

*If  $\Omega$  is a type domain of  $A$ , then*

1.  $L^{p,0}(\Omega) = L^p(\Omega)$
2.  $L^{p,n}(\Omega) = L^\infty(\Omega)$
3. For  $\mu > n$ ,  $L^{p,\mu}(\Omega) = 0$
4. If  $p \leq q$ ,  $\frac{n-\mu}{p} \geq \frac{n-\nu}{q}$ , then  $L^{q,\nu}(\Omega) \subset L^{p,\mu}(\Omega)$

**Lemma 4.10.** *Assume  $u \in H^1(\Omega)$ , satisfying  $\mu \in [0, n)$ , there exists constant  $M > 0$ , such that*

$$\int_{B_r(x_0)} |Du|^2 \leq Mr^\mu, \forall B_r(x_0) \subset \Omega$$

*Then for any  $\Omega' \subset \subset \Omega$ ,  $x_0 \in \Omega'$ ,  $B_r(x_0) \subset \Omega$ , we have*

$$\int_{B_r(x_0)} |u|^2 \leq c(n, \lambda, \mu, \Omega', \Omega) \left\{ M + \int_{\Omega} u^2 \right\} r^\lambda$$

where

$$\lambda = \begin{cases} \mu + 2, & \mu < n - 2 \\ \text{any number in } [0, n), & n - 2 \leq \mu < n \end{cases}$$

*Proof.* Let  $R_0 = \text{dist}(\Omega', \partial\Omega)$ , then for any  $x_0 \in \Omega'$ ,  $0 < r \leq R_0$ , we have by Poincaré inequality,

$$\int_{B_r(x_0)} |u - u_{x_0,r}|^2 \leq c(n)Mr^{\mu+2}$$

Then for  $\lambda$  defined in the lemma, we have

$$\int_{B_r(x_0)} |u - u_{x_0,r}|^2 \leq c(n)Mr^\lambda$$

i.e.  $u \in \mathcal{L}^{2,\mu}(\Omega')$ .

For any  $0 < \rho < r \leq R_0$ ,

$$\begin{aligned} \int_{B_\rho(x_0)} |u|^2 &\leq 2 \int_{B_\rho(x_0)} |u - u_{x_0,r}|^2 + 2 \int_{B_\rho(x_0)} |u_{x_0,r}|^2 \\ &\leq c(n)\rho^n |u_{x_0,r}|^2 + c(n)Mr^\lambda \\ &\leq c(n) \left\{ \left( \frac{\rho}{r} \right)^n \int_{B_r(x_0)} |u|^2 + Mr^\lambda \right\} \end{aligned}$$

Let  $\phi(r) = \int_{B_r(x_0)} |u|^2$ , then

$$\phi(\rho) \leq c(n) \left\{ \left( \frac{\rho}{r} \right)^n \phi(r) + Mr^\lambda \right\}$$

Using the lemma below, we can get the desired result. □

**Lemma 4.11.** *Assume  $\phi(r)$  is a non-decreasing and non-negative function in  $[0, R]$ , satisfying, for  $0 < \rho < r \leq R$ ,*

$$\phi(\rho) \leq A \left\{ \left( \frac{\rho}{r} \right)^\alpha + \varepsilon \right\} \phi(r) + Br^\beta$$

where  $A, \alpha, B, \beta \geq 0, \alpha > \beta$ . Then for any  $\gamma \in (\beta, \alpha)$ , there exists  $\varepsilon_0 = \varepsilon_0(A, \alpha, B, \beta, r)$ , such that if  $\varepsilon < \varepsilon_0$ , we have

$$\phi(\rho) \leq c \left\{ \left( \frac{\rho}{r} \right)^\gamma \phi(r) + B\rho^\beta \right\}$$

where  $c = c(\alpha, \beta, \gamma, A)$ . In particular,

$$\phi(r) = c \left\{ \left( \frac{r}{R} \right)^\gamma \phi(R) + Br^\beta \right\}$$

**Remark 4.12.** From the lemma,  $\mu \in [0, n)$ , we have  $\mathcal{L}_{loc}^{p,\mu}(\Omega) = L_{loc}^{p,\mu}(\Omega)$ . Even if the proof is based on  $p = 2$ , the result is true for any  $p \geq 1$ .

We also remark that the lemma deals with the local situation. Through a similar argument, we can prove  $\mathcal{L}^{p,\mu}(\Omega) = L^{p,\mu}(\Omega)$  for any  $p \geq 1$  and  $\mu \in [0, n)$ .

### 4.3 John-Nirenberg Theorem and BMO space

**Theorem 4.13** (John-Nirenberg). Assume  $u \in L^1(\Omega)$ , satisfying

$$\int_{B_r(x_0)} |u - u_{x_0,r}| \leq Mr^n, \forall B_r(x_0) \subset \Omega, M > 0$$

Then, for any  $B_r(x_0) \subset \Omega$ , there holds

$$\int_{B_r(x_0)} e^{\frac{p_0}{M}|u - u_{x_0,r}|} \leq Cr^n$$

for some positive  $p_0$  and  $C = C(n)$

The proof relies on the Calderon-Zygmund decomposition in harmonic analysis.

**Remark 4.14.** 1. For  $Q_0 \subset \mathbb{R}^n$  a cube and  $u \in L^1(Q_0)$ , if the seminorm

$$|u|_{*,Q} = \sup_{Q \subset Q_0} \frac{1}{|Q|} \int_Q |u - u_{Q,r}| < \infty$$

then we call  $u$  is a function of bounded mean oscillation (BMO). And the norm of BMO space is defined as

$$\|u\|_{BMO(\Omega)} = \|u\|_{L^1(Q_0)} + |u|_{*,Q_0}$$

The BMO space under such norm is a Banach space. We note that we let the domain to be the cube for the convenience of technique in harmonic analysis.

2.  $\mathcal{L}^{1,n}(Q_0) = BMO(Q_0)$ . Furthermore,  $\mathcal{L}^{p,n}(Q_0) = BMO(Q_0)$  by the John-Nirenberg theorem.
3.  $L^\infty(Q_0) \subsetneq BMO(Q_0)$ .

Now we have classified the Campanato space with all the index  $\mu$ .

1.  $\mathcal{L}^{p,\mu}(\Omega) = L^{p,\mu}(\Omega)$  for  $\mu \in [0, n)$  (by lemma 4.10).
2.  $\mathcal{L}^{p,n}(\Omega) = BMO(\Omega)$  by theorem 4.13.
3.  $\mathcal{L}^{p,\mu}(\Omega) = C^\alpha(\bar{\Omega})$  for  $n < \mu \leq n + p, \alpha = \frac{\mu - n}{p}$  (by theorem 4.1).
4.  $\mathcal{L}^{p,\mu}(\Omega) = \{\text{const}\}$  for  $\mu > n + p$ .

## 5 Regularity of Solutions

### 5.1 Hölder continuity of solutions

#### 5.1.1 Basic estimates of harmonic functions

**Main ideas:**

1. Freeze the coefficients.
2. Compare with harmonic functions. (The regularity of solutions depends on how closed the solution is to harmonic functions.)

Now we introduce the first lemma.

**Lemma 5.1.** *Suppose  $\{a^{ij}\}$  is a constant positive definite matrix, and  $w \in H^1(\Omega)$  is a solution to the linear elliptic equation*

$$a^{ij} D_{ij} w = 0.$$

*Then for any  $B_r(x_0) \subset\subset \Omega$  and  $0 < \rho \leq r$  we have*

$$\int_{B_\rho(x_0)} |w|^2 \leq C \left( \frac{\rho}{r} \right)^n \int_{B_r(x_0)} |w|^2$$

*and*

$$\int_{B_\rho(x_0)} |w - w_{x_0, \rho}|^2 \leq C \left( \frac{\rho}{r} \right)^n \int_{B_r(x_0)} |w - w_{x_0, r}|^2$$

*We note that the estimates also hold if we change  $w$  into  $Dw$ .*

*Proof.* 1. We first prove the Caccioppoli inequality. For any  $\eta \in C_0^1(B_r(x_0))$ , we choose  $\phi = \eta^2 w$ , then by the definition of weak solution, we have

$$\int_{B_r(x_0)} a^{ij} D_i w D_j w \eta^2 = -2 \int_{B_r(x_0)} a^{ij} w D_i w D_j \eta \eta$$

Using the uniform ellipticity and Hölder inequality, we have

$$\int_{B_r(x_0)} \eta^2 |Dw|^2 \leq c \int_{B_r(x_0)} |D\eta|^2 |w|^2$$

For  $0 < \rho < r$ , choose  $\eta \equiv 1$  in  $B_\rho(x_0)$ ,

$$\int_{B_\rho(x_0)} |Dw|^2 \leq \frac{c}{(r - \rho)^2} \int_{B_r(x_0)} |w|^2$$

which implies the Caccioppoli inequality.

2. **Claim:** For  $0 < \rho < r$ , we have

$$\int_{B_{\rho/2}} |w|^2 \leq \theta \int_{B_\rho(x_0)} |w|^2$$

and

$$\int_{B_{\rho/2}} |Dw|^2 \leq \theta \int_{B_\rho(x_0)} |Dw|^2$$

for some  $\theta = \theta(n, \lambda, \Lambda) \in (0, 1)$ .

We will use a "Hole-filling" argument.

In fact, choose  $\eta \in C_0^1(B_\rho(x_0))$ ,  $\eta \equiv 1$  in  $B_{\rho/2}(x_0)$ , by the Caccioppoli inequality, we have

$$\int_{B_\rho(x_0)} |D(\eta w)|^2 \leq C \int_{B_\rho(x_0)} |D\eta|^2 |w|^2 \leq \frac{C}{R^2} \int_{B_\rho(x_0) - B_{\rho/2}(x_0)} |w|^2$$

By the Poincaré inequality,

$$\int_{B_{\rho/2}(x_0)} |w|^2 \leq C \int_{B_\rho(x_0) - B_{\rho/2}(x_0)} |w|^2$$

Filling the hole we get

$$\int_{B_{\rho/2}(x_0)} |w|^2 \leq \frac{C}{C+1} \int_{B_\rho(x_0)} |w|^2$$

To prove the second inequality, we only to notice that the Caccioppoli inequality also holds for  $w - a$ , then the same argument shows that

$$\int_{B_\rho} \eta^2 |Dw|^2 \leq \frac{C}{R^2} \int_{B_\rho(x_0) - B_{\rho/2}(x_0)} |w - a|^2$$

We choose  $a$  to be the average of  $w$  and by Poincaré inequality,

$$\int_{B_{\rho/2}(x_0)} |Dw|^2 \leq C \int_{B_\rho(x_0) - B_{\rho/2}(x_0)} |Dw|^2$$

Through a filling-hole argument we get the desired inequality. □

**Proposition 5.2.** *Suppose  $u$  satisfies*

$$a^{ij} D_{ij} u = 0$$

*Then for any  $v \in H^1(B_1(0))$ , there holds, for  $0 < \rho < r$ ,  $B_r(x) \subset B_1(0)$ ,*

$$\int_{B_\rho(x)} |Dv|^2 \leq C \left\{ \left( \frac{\rho}{r} \right)^n \int_{B_r(x)} |Dv|^2 + \int_{B_r(x)} |D(u - v)|^2 \right\}$$

*and*

$$\int_{B_\rho(x)} |Dv - (Dv)_{x,\rho}|^2 \leq C \left\{ \left( \frac{\rho}{r} \right)^n \int_{B_r(x)} |Dv - (Dv)_{x,r}|^2 + \int_{B_r(x)} |D(u - v)|^2 \right\}$$

*where  $C = C(\lambda, \Lambda)$*

*Proof.* Let  $w = v - u$ , then

$$\int_{B_\rho(x)} |Dv|^2 \leq 2 \left( \int_{B_\rho(x)} |Dw|^2 + \int_{B_\rho(x)} |Du|^2 \right)$$

Then apply lemma 5.1 to get the desired inequality. □

### 5.1.2 Hölder Continuity

**Theorem 5.3.** Assume that  $a^{ij}$  is uniformly elliptic in  $B_1(0)$ , and  $a^{ij} \in C(\bar{B}_1(0))$ . And  $u \in H^1(B_1)$  satisfies the equation (divergence form) in the weak sense, i.e.

$$\int_{B_1} (a^{ij} D_i u D_j \phi + cu\phi) = \int_{B_1} f\phi$$

for all  $\phi \in H_0^1(B_1)$ . And  $c \in L^n(B_1)$ ,  $f \in L^q(B_1)$  with  $q \in (\frac{n}{2}, n)$ .

Then  $u \in C^\alpha(\bar{B}_1(0))$  with  $\alpha = 2 - \frac{n}{q}$ .

Furthermore, there exists  $R_0 = R_0(\lambda, \Lambda, \|c\|_{L^n(B_1)}, \tau) > 0$ , such that for any  $0 < r < R_0 < 1$ ,  $x \in B_{1/2}$ , we have

$$\int_{B_r(x)} |Du|^2 \leq Cr^{n-2+2\alpha} \left\{ \|f\|_{L^q(B_1(x))}^2 + \|u\|_{H^1(B_1(x))}^2 \right\}$$

where  $\tau$  is the continuity modulus of  $a^{ij}$ .

*Proof.* 1. We can freeze the coefficients.

For any  $B_r(x_0) \subset B_1(0)$ , consider

$$\begin{cases} a^{ij}(x_0) D_{ij} w = 0 \\ w|_{\partial B_r(x_0)} = u \end{cases} \quad (2)$$

There exists a unique solution to the above equation.

Let  $v = w - u \in H_0^1(B_r(x_0))$ , then

$$\int_{B_r(x_0)} a^{ij}(x_0) D_i v D_j \phi = \int_{B_r(x_0)} f\phi - cu\phi + (a^{ij}(x_0) - a^{ij}(x)) D_i u D_j \phi$$

for any  $\phi \in H_0^1(B_r(x_0))$ .

Choose  $\phi = v$ , we have

$$\lambda \int_{B_r(x_0)} |Dv|^2 \leq \int_{B_r(x_0)} \tau(|x - x_0|) |Du| |Dv| + \int_{B_r(x_0)} |c| |u| |v| + \int_{B_r(x_0)} |f| |v|$$

We will use the Sobolev inequality

$$\left( \int_{B_r(x_0)} |v|^{\frac{2n}{n-2}} \right)^{\frac{n-2}{2n}} \leq c(n) \left( \int_{B_r(x_0)} |Dv|^2 \right)^{\frac{1}{2}}$$

for  $v \in H_0^1(B_r(x_0))$  (note that  $c$  is constant depending ONLY on  $n$ ).

Using the Hölder continuity to get the index  $\frac{2n}{n-2}$  on  $v$ , we can get

$$\begin{aligned} \int_{B_r(x_0)} |Dv|^2 &\leq c \left\{ \tau^2(r) \int_{B_r(x_0)} |Du|^2 + \left( \int_{B_r(x_0)} |c|^n \right)^{\frac{2}{n}} \int_{B_r(x_0)} u^2 \right. \\ &\quad \left. + \left( \int_{B_r(x_0)} |f|^{\frac{2n}{n+2}} \right)^{\frac{n+2}{n}} \right\} \end{aligned}$$

Then the proposition implies that for  $0 < \rho \leq r$ ,

$$\begin{aligned} \int_{B_\rho(x_0)} |Du|^2 &\leq c \left\{ \left[ \left( \frac{\rho}{r} \right)^n + \tau^2(r) \right] \int_{B_r(x_0)} |Du|^2 \right. \\ &\quad \left. + \|c\|_{L^n(B_r(x_0))}^2 \int_{B_r(x_0)} u^2 + \|f\|_{L^{\frac{2n}{n+2}}(B_r(x_0))}^2 \right\} \end{aligned}$$

Using Hölder inequality,

$$\left( \int_{B_r(x_0)} |f|^{\frac{2n}{n+2}} \right)^{\frac{n+2}{n}} \leq c \left( \int_{B_r(x_0)} |f|^q \right)^{\frac{2}{q}} r^{n-2+2\alpha}$$

Then we have

$$\begin{aligned} \int_{B_\rho(x_0)} |Du|^2 \leq C \left\{ \left[ \left( \frac{\rho}{r} \right)^n + \tau^2(r) \right] \int_{B_r(x_0)} |Du|^2 \right. \\ \left. + \|c\|_{L^n(B_r(x_0))}^2 \int_{B_r(x_0)} u^2 + r^{n-2+2\alpha} \|f\|_{L^q(B_r(x_0))}^2 \right\} \end{aligned} \quad (3)$$

2. Consider the special case  $c \equiv 0$ . Then equation (3) becomes

$$\begin{aligned} \int_{B_\rho(x_0)} |Du|^2 \leq C \left\{ \left[ \left( \frac{\rho}{r} \right)^n + \tau^2(r) \right] \int_{B_r(x_0)} |Du|^2 \right. \\ \left. + r^{n-2+2\alpha} \|f\|_{L^q(B_r(x_0))}^2 \right\}. \end{aligned} \quad (4)$$

Using the iteration lemma in Lecture 15, we get that for  $R_0$  sufficiently small ( $\tau^2(r) < \varepsilon_0$  in the iteration lemma), and  $0 < \rho < R_0$ ,

$$\int_{B_\rho(x_0)} |Du|^2 \leq C \rho^{n-2+2\alpha} \left\{ \int_{B_1} |Du|^2 + \|f\|_{L^q(B_1)}^2 \right\}$$

We conclude  $u \in C^\alpha(B_1)$  through the Morrey theorem.

3. **The general case** ( $c \not\equiv 0$ ). We now drop  $c \equiv 0$ . The inequality (3) already contains the lower-order terms; as in Gilbarg–Trudinger we compress the dependence on  $\|c\|_{L^n(B_1)}$  into the constant  $C$  and isolate the forcing energy by setting

$$\chi(F) := \|f\|_{L^q(B_1)}^2.$$

Then for any  $B_r(x_0) \subset B_1$  and any  $0 < \rho \leq r$ ,

$$\begin{aligned} \int_{B_\rho(x_0)} |Du|^2 \leq C \left\{ \left[ \left( \frac{\rho}{r} \right)^n + \tau^2(r) \right] \int_{B_r(x_0)} |Du|^2 \right. \\ \left. + r^{n-2+2\alpha} \chi(F) + \int_{B_r(x_0)} u^2 \right\}. \end{aligned} \quad (5)$$

We shall prove that for any  $x_0 \in B_{1/2}$  and any  $0 < \rho < r \leq \frac{1}{2}$ ,

$$\begin{aligned} \int_{B_\rho(x_0)} |Du|^2 \leq C \left\{ \left[ \left( \frac{\rho}{r} \right)^n + \tau^2(r) \right] \int_{B_r(x_0)} |Du|^2 \right. \\ \left. + r^{n-2+2\alpha} \left[ \chi(F) + \int_{B_1} u^2 + \int_{B_1} |Du|^2 \right] \right\}. \end{aligned} \quad (6)$$

This requires a bootstrap argument. First, by the iteration lemma in Lecture 15 (applied to  $u$ ), there exist  $R_1 \in (\frac{1}{2}, 1)$  and an exponent  $\delta_1 > 0$  such that for any  $x_0 \in B_{R_1}$  and any  $0 < r \leq 1 - R_1$ ,

$$\int_{B_r(x_0)} u^2 \leq C r^{\delta_1} \left\{ \int_{B_1} |Du|^2 + \int_{B_1} u^2 \right\}, \quad (7)$$

where  $\delta_1 = 2$  if  $n > 2$  and  $\delta_1$  is arbitrary in  $(0, 2)$  if  $n = 2$ .

Combining (7) with (5) gives

$$\int_{B_\rho(x_0)} |Du|^2 \leq C \left\{ \left[ \left( \frac{\rho}{r} \right)^n + \tau^2(r) \right] \int_{B_r(x_0)} |Du|^2 + r^{n-2+2\alpha} \chi(F) + r^{\delta_1} \|u\|_{H^1(B_1)}^2 \right\}.$$

Then (6) holds if either  $n = 2$  and we choose  $\delta_1 = 2\alpha$ , or  $n > 2$  and  $n - 2 + 2\alpha \leq 2$ , taking  $\delta_1 = 2$ .

For  $n > 2$  and  $n - 2 + 2\alpha > 2$ , we have instead

$$\int_{B_\rho(x_0)} |Du|^2 \leq C \left\{ \left[ \left( \frac{\rho}{r} \right)^n + \tau^2(r) \right] \int_{B_r(x_0)} |Du|^2 + r^2 [\chi(F) + \|u\|_{H^1(B_1)}^2] \right\}.$$

Applying the same iteration scheme to  $|Du|$  yields, for any  $x_0 \in B_{R_1}$  and any  $0 < r \leq 1 - R_1$ ,

$$\int_{B_r(x_0)} |Du|^2 \leq Cr^2 \{ \chi(F) + \|u\|_{H^1(B_1)}^2 \}.$$

Hence there exists  $R_2 \in (\frac{1}{2}, R_1)$  such that for any  $x_0 \in B_{R_2}$  and any  $0 < r \leq R_1 - R_2$ ,

$$\int_{B_r(x_0)} u^2 \leq Cr^{\delta_2} \{ \chi(F) + \|u\|_{H^1(B_1)}^2 \}, \quad (8)$$

where  $\delta_2 = 4$  if  $n > 4$  and  $\delta_2$  is arbitrary in  $(2, n)$  if  $n = 3$  or  $4$ . Estimate (8) improves (7). Substituting (8) into (5) and repeating this finitely many times, we obtain (6). With (6) and the iteration lemma in Lecture 15, the same Campanato–Morrey argument as in the case  $c \equiv 0$  yields  $u \in C^\alpha(B_{1/2})$  (hence on compact subsets of  $B_1$  by covering). This finishes the proof.

The process above is called the **Bootstrap Argument**.

□

## 6 $L^p$ Theory for Elliptic Equations

### 6.1 Interpolation Theorem

**Definition 6.1** (Distribution function). Let  $(X, \mu)$  be a measure space, and  $f$  is a measurable function on  $X$ . Then the distribution function of  $f$  is defined as

$$D_f : [0, +\infty) \rightarrow [0, +\infty), \quad t \mapsto \mu(\{x \in X : |f(x)| > t\})$$

**Lemma 6.2** (Slice lemma). If  $f \in L^p(X, \mu)$ ,  $0 < p < +\infty$ , then

$$\|f\|_{L^p} = p \int_0^{+\infty} t^{p-1} D_f(t) dt$$

Then we can define the weak  $L^p$  space  $L_w^p(X, \mu)$ :

**Definition 6.3.** The weak space  $L_w^p(X, \mu)$  is defined as the set of all  $\mu$ -measurable functions  $f$  such that

$$\|f\|_{L_w^p(X, \mu)} := \inf \left\{ A > 0 : D_f(t) \leq \left(\frac{A}{t}\right)^p, \forall t > 0 \right\} < \infty$$

*Remark 6.4.* Let  $X$  be a finite measure space, then

1.  $L_w^\infty(X, \mu) = L^\infty(X, \mu)$
2.  $L^p(X, \mu) \subsetneq L_w^p(X, \mu) \subset L^q(X, \mu)$  for  $1 \leq q < p < +\infty$

**Definition 6.5.** Let  $\Omega$  be a bounded domain in  $\mathbb{R}^n$ . We call  $T : L^p(\Omega) \rightarrow L^q(\Omega)$  a quasilinear mapping if there exists a constant  $Q > 0$  such that, for any  $f, g \in L^p(\Omega)$ ,

$$|T(f+g)| \leq Q(|T(f)| + |T(g)|) \quad \text{a.e. in } \Omega$$

Furthermore, we call the quasilinear mapping  $T$  is strong type of  $(p, q)$  if there exists a constant  $C > 0$  such that, for any  $f \in L^p(\Omega)$ ,

$$\|Tf\|_{L^q(\Omega)} \leq C\|f\|_{L^p(\Omega)}$$

Denote the norm of  $T$  as

$$\|T\|_{(p,q)} = \sup_{f \in L^p(\Omega), f \neq 0} \frac{\|Tf\|_{L^q(\Omega)}}{\|f\|_{L^p(\Omega)}}.$$

Similarly, we can define the weak type of  $(p, q)$  operator  $T$ , if there exists a constant  $C > 0$  such that, for any  $f \in L^p(\Omega)$ ,

$$\|Tf\|_{L_w^q(\Omega)} \leq C\|f\|_{L^p(\Omega)}.$$

*Remark 6.6.* It is easy to verify that "strong type" implies "weak type".

**Theorem 6.7** (Marcinkiewicz Inner Interpolation Theorem). Assume  $1 \leq p < q \leq \infty$ , and the quasilinear mapping  $T$  is weak type of  $(p, p)$  and weak type of  $(q, q)$ , i.e.

$$\|Tf\|_{L_w^p(\Omega)} \leq B_p\|f\|_{L^p(\Omega)}$$

and

$$\|Tg\|_{L_w^q(\Omega)} \leq B_q\|g\|_{L^q(\Omega)}$$

for some constants  $B_p, B_q > 0$ .

Then for any  $r$  such that  $p < r < q$ , the quasilinear mapping  $T$  is strong type of  $(r, r)$ , with the estimate

$$\|T\|_{r,r} \leq C(B_p^\theta B_q^{1-\theta})$$

where  $C = C(r, p, q, Q)$  and  $\theta = \frac{p}{r} \frac{q-r}{q-p}$ .

*Proof.* 1. If  $f \in L^r(\Omega)$ , we write  $f = f_1 + f_2$  where

$$f_1 = \begin{cases} f, & |f| \leq \gamma s \\ 0, & |f| > \gamma s \end{cases}$$

and

$$f_2 = \begin{cases} 0, & |f| \leq \gamma s \\ f, & |f| > \gamma s \end{cases}$$

with  $\gamma$  a constant to be fixed. Then  $f_1 \in L^q(\Omega)$  and  $f_2 \in L^p(\Omega)$ .

Since  $T$  is quasilinear,

$$\{x \in \Omega : |Tf(x)| > s\} \subset \left\{x \in \Omega : |Tf_1(x)| > \frac{s}{2Q}\right\} \cup \left\{x \in \Omega : |Tf_2(x)| > \frac{s}{2Q}\right\}$$

which implies

$$D_{Tf}(s) \leq D_{Tf_1}(s/2Q) + D_{Tf_2}(s/2Q)$$

Since  $f_1 \in L^q(\Omega)$ , we have

$$\|Tf_1\|_{L_w^q(\Omega)} \leq B_q \|f_1\|_{L^q(\Omega)}$$

According to the definition, we have

$$\frac{s}{2Q} D_{Tf_1} \left( \frac{s}{2Q} \right)^{1/q} \leq B_q \|f_1\|_{L^q(\Omega)}$$

i.e.

$$D_{Tf_1}(s/2Q) \leq \left( \frac{2QB_q}{s} \right)^q \|f_1\|_{L^q(\Omega)}^q$$

Similarly, we have

$$D_{Tf_2}(s/2Q) \leq \left( \frac{2QB_p}{s} \right)^p \|f_2\|_{L^p(\Omega)}^p$$

Therefore,

$$D_{Tf}(s) \leq \left( \frac{2QB_q}{s} \right)^q \|f_1\|_{L^q(\Omega)}^q + \left( \frac{2QB_p}{s} \right)^p \|f_2\|_{L^p(\Omega)}^p.$$

2. Using lemma 6.2,

$$\begin{aligned} \int_{\Omega} |Tf|^r &\leq \int_0^\infty r s^{r-1} D_{Tf}(s) ds \\ &\leq (2QB_p)^p \int_0^\infty r s^{r-p-1} ds \int_{|f|>\gamma s} |f|^p ds + (2QB_q)^q \int_0^\infty r s^{r-q-1} ds \int_{|f|>\gamma s} |f|^q ds \\ &= (2QB_p)^p r \int_{\Omega} |f|^p dx \int_0^{|f|/\gamma} s^{r-p-1} ds + (2QB_q)^q r \int_{\Omega} |f|^q dx \int_0^{|f|/\gamma} s^{r-q-1} ds \\ &= \left[ \frac{(2QB_p)^p r}{r-p} \gamma^{p-r} + \frac{(2QB_q)^q r}{q-r} \gamma^{q-r} \right] \int_{\Omega} |f|^r dx \end{aligned}$$

Choosing  $\gamma = (B_p^p B_q^{-q})^{\frac{1}{q-p}}$  we get

$$\|T\|_{(r,r)} \leq C B_p^\theta B_q^{1-\theta}$$

3. If  $q = +\infty$ , choose  $\gamma$  large enough and use similar arguments as above.

□

## 6.2 $L^p$ estimates for Poisson Equation

Consider

$$\Delta u = 0 \text{ in } \mathbb{R}^n, \quad n \geq 3$$

and

$$\Gamma(x) = \frac{1}{n(2-n)\omega_n} \frac{1}{|x|^{n-2}}$$

For  $f \in C_0^\infty(\mathbb{R}^n)$ , we define the Newtonian potential of  $f$  as

$$w(x) = - \int_{\mathbb{R}^n} \Gamma(x - \xi) f(\xi) d\xi$$

**Lemma 6.8.** *If  $f \in C_0^\infty(\mathbb{R}^n)$ , then  $w(x) \in C_0^\infty(\mathbb{R}^n)$  and*

$$-\Delta w = f \text{ in } \mathbb{R}^n.$$

We define the operator such that  $w = Nf$  where

$$N : C_0^\infty(\mathbb{R}^n) \rightarrow C^\infty(\mathbb{R}^n)$$

which is linear. And for fixed  $i, j$ , consider

$$Tf := D_{ij}Nf.$$

We will derive a  $W^{2,p}$  estimate for  $Tf$ .

**Lemma 6.9.**  *$T$  is a bounded linear operator from  $L^2(\mathbb{R}^n)$  to  $L^2(\mathbb{R}^n)$ , and  $\|T\|_{(2,2)} \leq 1$ .*

*Proof.* 1. Assume  $f \in C_0^\infty(\mathbb{R}^n)$ , and  $\text{supp}(f) \subset B_{R_0}$ , then for any  $R > 2R_0$ , we have

$$\begin{aligned} \int_{B_R} f^2 &= \int_{B_R} f \Delta w = \int_{\partial B_R} f \frac{\partial w}{\partial \nu} - \int_{B_R} Df \cdot Dw \\ &= - \int_{B_R} \Delta(Dw) \cdot Dw = \sum_{i=1}^n \int_{B_R} \Delta w_i w_i \\ &= \int_{B_R} \sum_{i,j=1}^n |D^2 w|^2 - \int_{\partial B_R} Dw \cdot \frac{\partial}{\partial \nu} Dw \end{aligned}$$

For  $x \in \partial B_R$ , we have

$$|D_i w| \leq \int_{B_{R_0}} |D_i \Gamma(x - \xi)| |f(\xi)| d\xi \leq \frac{C}{R^{n-1}}$$

and

$$|D^2 w| \leq \frac{C}{R^n}$$

Let  $R \rightarrow \infty$  we get that

$$\int_{\mathbb{R}^n} f^2 = \int_{\mathbb{R}^n} \sum_{i,j=1}^n (D_{ij}w)^2$$

2. To deal with the general case, we can prove by approximation: Let  $\{f_n\} \subset C_0^\infty(\mathbb{R}^n)$  is a sequence of smooth functions that tends to  $f$  in  $L^2(\mathbb{R}^n)$ , and we can define  $w_n = Nf_n$ . Then by the argument above,

$$\|D^2 w_m - D^2 w_n\|_{L^2(\mathbb{R}^n)} = \|f_m - f_n\|_{L^2(\mathbb{R}^n)}.$$

Hence  $\{D^2 w_n\}$  is a Cauchy sequence in  $L^2(\mathbb{R}^n)$ .

Then  $T$  naturally extends to a bounded linear operator from  $L^2(\mathbb{R}^n)$  to  $L^2(\mathbb{R}^n)$ . (Note that it is equivalent to replace  $\mathbb{R}^n$  by any domain, since we can always use zero extension.)  $\square$

Next our target is to prove that  $T$  is weak type of  $(1, 1)$ , then by Marcinkiewicz Interpolation we conclude that  $T$  is strong type of  $(p, p)$ , for  $1 < p \leq 2$ . Then by a duality argument  $T$  is strong type of  $(p, p)$  for  $1 < p < \infty$ .

**Lemma 6.10.** *For  $\Gamma(x)$ , we define*

$$J := \sup_{\xi \neq 0, 1 \leq i, j \leq n} \int_{|x| \geq 2|\xi|} |D_{ij}\Gamma(x - \xi) - D_{ij}\Gamma(x)| dx.$$

Then  $J \leq C(n) < \infty$ .

*Proof.* The kernel  $D_{ij}\Gamma$  is homogeneous of degree  $-n$ , and its gradient is bounded by  $C|x|^{-n-1}$ . If  $|x| \geq 2|\xi|$ , then  $|x - t\xi| \geq |x|/2$  for  $0 \leq t \leq 1$ . By the differential mean value theorem,

$$|D_{ij}\Gamma(x - \xi) - D_{ij}\Gamma(x)| \leq C|\xi||x|^{-n-1}.$$

Therefore

$$\int_{|x| \geq 2|\xi|} |D_{ij}\Gamma(x - \xi) - D_{ij}\Gamma(x)| dx \leq C|\xi| \int_{2|\xi|}^{\infty} r^{-2} dr \leq C.$$

Taking the supremum proves the result.  $\square$

**Theorem 6.11.**  *$T$  is weak type of  $(1, 1)$ .*

*Proof.* 1. First assume that  $f \in C_0^\infty(\mathbb{R}^n)$ , by the Calderon-Zygmund decomposition theorem, we have for  $|f(x)|$  and fixed  $\alpha > 0$ , there exists two sets  $F, \Omega$  such that  $\mathbb{R}^n = F \cup \Omega$  and,  $|f(x)| \leq \alpha$  a.e. in  $F$  and  $\Omega = \cup_{i=1}^{\infty} Q_i$  ( $Q_i$  are disjoint cubes), and

$$|\Omega| = \sum_{i=1}^{\infty} |Q_i| \leq \frac{1}{\alpha} \|f\|_{L^1(\mathbb{R}^n)}, \quad \frac{1}{|Q_i|} \int_{Q_i} |f| \leq 2^n \alpha.$$

Define

$$g(x) = \begin{cases} f(x), & x \in F \\ \frac{1}{|Q_i|} \int_{Q_i} f(y) dy, & x \in Q_i \end{cases}$$

2. Then we choose  $h(x) = f(x) - g(x)$ , then we have

$$\|g\|_{L^1(\mathbb{R}^n)} \leq \|f\|_{L^1(\mathbb{R}^n)}, \quad \|h\|_{L^1(\mathbb{R}^n)} \leq 2\|f\|_{L^1(\mathbb{R}^n)}$$

Then  $h(x) = 0$  if  $x \in F$  and  $\int_{Q_i} h(x) dx = 0$  for all  $i$ .

Since  $f \in C_0^\infty(\mathbb{R}^n)$ ,  $g, h \in L^2(\mathbb{R}^n)$ , and  $Tf = Tg + Th$ ,

$D_{Tf}(\alpha) \leq D_{Tg}(\alpha/2) + D_{Th}(\alpha/2)$ , then we estimate the two terms separately:

3. For  $D_{Tg}(\alpha/2)$ ,

$$D_{Tg}(\alpha/2) \leq \frac{4}{\alpha^2} \|g\|_{L^2(\mathbb{R}^n)}^2 \leq \frac{4}{\alpha^2} \|g\|_{L^\infty(\mathbb{R}^n)} \|g\|_{L^1(\mathbb{R}^n)} \leq \frac{2^{n+2}}{\alpha} \|f\|_{L^1(\mathbb{R}^n)}$$

4. For  $D_{Th}(\alpha/2)$ , we first enlarge the cube  $Q_i$  to  $Q_i^*$  such that  $Q_i^* = 2\sqrt{n}Q_i$ , and denote  $\Omega^* = \cup_{i=1}^{\infty} Q_i^*$ ,  $F^* = \mathbb{R}^n \setminus \Omega^*$ , then

$$|\Omega^*| \leq (2\sqrt{n})^n |\Omega| \leq \frac{(2\sqrt{n})^n}{\alpha} \|f\|_{L^1(\mathbb{R}^n)}$$

We use a cut-off argument, define  $h_k(x) = h(x)\chi_{Q_k}$  which supports in  $Q_k$ .

We choose a sequence of smooth functions  $\{h_k^l\} \subset C_0^\infty(Q_k)$  such that

$$\|h_k^l - h_k\|_{L^2(Q_k)} \rightarrow 0 \text{ as } l \rightarrow \infty, \quad \int_{Q_k} h_k^l(x) dx = 0$$

Then for  $x \in (Q_k^*)^c$ , we have

$$\begin{aligned} Th_k^l(x) &= D_{ij} \int_{Q_k} \Gamma(x - \xi) h_k^l(\xi) d\xi \\ &= D_{ij} \int_{Q_k} (\Gamma(x - \xi) - \Gamma(x - x^k)) h_k^l(\xi) d\xi \quad (x^k \text{ is the center of } Q_k) \end{aligned}$$

Integrating over  $(Q_k^*)^c$  we get

$$\begin{aligned} \int_{(Q_k^*)^c} |Th_k^l(x)| dx &\leq \int_{(Q_k^*)^c} \left| D_{ij} \int_{Q_k} (\Gamma(x - \xi) - \Gamma(x - x^k)) h_k^l(\xi) d\xi \right| dx \\ &\leq \sup_{\xi} \int_{(Q_k^*)^c} |D_{ij} \Gamma(x - \xi) - D_{ij} \Gamma(x - x^k)| dx \int_{Q_k} |h_k^l(\xi)| d\xi \\ &\leq J \int_{Q_k} |h_k^l(\xi)| d\xi \end{aligned}$$

By Fatou lemma and the strong (2, 2) type property of  $T$ , we get

$$\int_{(Q_k^*)^c} |Th_k(x)| dx \leq J \int_{Q_k} |h_k(\xi)| d\xi = J \int_{Q_k} |h(\xi)| d\xi$$

Therefore,

$$\int_{F^*} |Th(x)| dx \leq J \|h\|_{L^1(\mathbb{R}^n)} \leq 2J \|f\|_{L^1(\mathbb{R}^n)}$$

Then

$$|\{x \in F^* : |Th(x)| \geq \alpha/2\}| \leq \frac{4}{\alpha} J \|f\|_{L^1(\mathbb{R}^n)}$$

combining the estimates,

$$D_{Th}(\alpha/2) \leq \frac{4}{\alpha} J \|f\|_{L^1(\mathbb{R}^n)} + |\Omega^*| \leq \frac{1}{\alpha} ((2\sqrt{n})^n + 4J) \|f\|_{L^1(\mathbb{R}^n)}$$

which implies

$$D_{Tf}(\alpha) \leq \frac{1}{\alpha} C(n) \|f\|_{L^1(\mathbb{R}^n)}$$

which implies that  $T$  is weak type of (1, 1) for smooth functions.

5. For the general case, using smooth approximation. □

**Theorem 6.12.** For  $1 < p < \infty$ ,  $T$  is strong type of  $(p, p)$ .

*Proof.* By Marcinkiewicz Interpolation Theorem, we have that  $T$  is strong type of  $(p, p)$  for  $1 < p \leq 2$ .

For  $2 < p < \infty$ , let  $p' = \frac{p}{p-1}$ , then for any  $f, h \in C_0^\infty(\mathbb{R}^n)$ ,

$$\begin{aligned} \int_{\mathbb{R}^n} Tfh &= \int_{\mathbb{R}^n} D_{ij} h(x) \int_{\mathbb{R}^n} \Gamma(x - \xi) f(\xi) d\xi dx \\ &= \int_{\mathbb{R}^n} f(\xi) \int_{\mathbb{R}^n} D_{ij} h(x) \Gamma(x - \xi) dx d\xi \\ &= \int_{\mathbb{R}^n} f(\xi) D_{ij} \int_{\mathbb{R}^n} \Gamma(x - \xi) h(x) dx d\xi \\ &= \|f\|_{L^p(\mathbb{R}^n)} \|Th\|_{L^{p'}(\mathbb{R}^n)} \leq C \|f\|_{L^p(\mathbb{R}^n)} \|h\|_{L^{p'}(\mathbb{R}^n)} \end{aligned}$$

which implies that  $T$  is strong type of  $(p, p)$  for  $2 < p < \infty$ . □

### 6.3 Calderon-Zygmund Estimates in $L^p$ cases

#### 6.3.1 The interior estimate

Using an approximation argument, we can prove that:

**Theorem 6.13.** *Assume  $u \in W_0^{2,p}(\Omega)$ ,  $-\Delta u = f$  in  $\Omega$ , then for  $1 < p < \infty$ , there exists  $c > 0$  such that*

$$\|D^2 u\|_{L^p(\Omega)} \leq c \|f\|_{L^p(\Omega)}$$

Then we consider the  $W^{2,p}$  estimate of the general linear elliptic equation.  
Consider the elliptic operator

$$Lu = -a^{ij}D_{ij}u + b^iD_iu + cu = f \text{ in } \Omega$$

and assume that

1.  $a^{ij}\xi_i\xi_j \geq \lambda|\xi|^2$  for  $\lambda > 0$ .
2.  $\sum_{i,j} \|a^{ij}\|_{L^\infty} + \sum_i \|b^i\|_{L^\infty} + \|c\|_{L^\infty} < \Lambda < \infty$
3.  $a^{ij} \in C(\bar{\Omega})$

**Lemma 6.14.** *Assume (1)(2)(3), and  $u \in W_0^{2,p}(B_R)$  with  $B_R \subset \subset \Omega$ ,  $1 < p < \infty$ , satisfying the linear elliptic equation. Then there exists  $R_0 = R_0(n, p, \Lambda, \lambda, \tau(a^{ij}))$  such that  $0 < R \leq R_0$ , there holds*

$$\|D^2 u\|_{L^p(B_R)} \leq \hat{C} \left\{ \frac{1}{\lambda} \|f\|_{L^p(B_R)} + R^{-2} \|u\|_{L^p(B_R)} \right\}$$

where  $\hat{C} = \hat{C}(n, \lambda, \Lambda)$

*Proof.* We will use the method of freezing the coefficients.

Let  $B_R = B_R(x_0)$ , then we can covert the equation in  $B_R$  to

$$-a^{ij}(x_0)D_{ij}u = \tilde{f}$$

where  $\tilde{f} = f + (a^{ij}(x) - a^{ij}(x_0))D_{ij}u - b^iD_iu - cu$ . Similar to the Poisson equation, we have that

$$\|D^2 u\|_{L^p(B_R)} \leq \hat{C} \|\tilde{f}\|_{L^p(B_R)}$$

where  $\hat{C} = \hat{C}(n, p, \lambda, \Lambda)$ .

Note that

$$\|\tilde{f}\|_{L^p(B_R)} \leq \|f\|_{L^p(B_R)} + \tau(R) \|D^2 u\|_{L^p(B_R)} + \Lambda \|u\|_{W^{1,p}(B_R)}$$

Then by the Sobolev interpolation inequality, we can get the desired estimate. □

**Theorem 6.15.** *Assume (1)(2)(3), and  $u \in W_{loc}^{2,p}(\Omega)$ , satisfying the linear elliptic equation. Then for any  $\Omega' \subset \subset \Omega$ , there holds*

$$\|u\|_{W^{2,p}(\Omega')} \leq \tilde{C} \left\{ \frac{1}{\lambda} \|f\|_{L^p(\Omega)} + \|u\|_{L^p(\Omega')} \right\}$$

where  $\tilde{C} = \tilde{C}(n, p, \lambda, \Lambda, \text{dist}(\Omega', \partial\Omega), \tau(a^{ij}))$ .

*Proof.* Choose  $\lambda = 1, R_0$  as stated in the previous lemma. Let  $\bar{R}_0 = \min R_0, \frac{1}{2} \text{dist}(\Omega', \partial\Omega)$ .

For any  $\frac{\bar{R}_0}{2} \leq \rho < R \leq \bar{R}_0, x_0 \in \Omega'$ , choose a cut-off function  $\zeta \in C_0^\infty(B_R(x_0))$  such that

$$\zeta \equiv 1 \text{ in } B_\rho(x_0), |D^k \zeta| \leq \frac{c}{(R - \rho)^k}, \quad k = 1, 2$$

Let  $v = u\zeta \in W_0^{2,p}(B_R)$ , then

$$Lv = \hat{f} := f\zeta + (-a^{ij}D_{ij}\zeta + b^iD_i\zeta)u - 2a^{ij}D_iuD_j\zeta.$$

We can now apply the previous lemma to get

$$\begin{aligned} \|D^2v\|_{L^p(B_R)} &\leq \tilde{C} \left\{ \|\hat{f}\|_{L^p(B_R)} + R^{-2}\|v\|_{L^p(B_R)} \right\} \\ &\leq \tilde{C} \left\{ \|f\|_{L^p(B_R)} + \frac{1}{(R - \rho)^2}\|u\|_{L^p(B_R)} + \frac{1}{(R - \rho)}\|Du\|_{L^p(B_R)} \right\} \end{aligned}$$

Then by Sobolev interpolation inequality,

$$\|D^2u\|_{L^p(B_\rho(x_0))} \leq \varepsilon \|D^2u\|_{L^p(B_R)} + \tilde{C} \left\{ \|f\|_{L^p(B_R(x_0))} + \frac{1}{(R - \rho)^2}\|u\|_{L^p(B_R)} \right\}$$

We come across the problem that the left hand side is  $B_\rho$  however the right hand side is  $B_R$ . We need an iteration result lemma 6.16 to acquire

$$\|D^2u\|_{L^p(B_\rho(x_0))} \leq C \left\{ \|f\|_{L^p(B_R(x_0))} + \frac{1}{(R - \rho)^2}\|u\|_{L^p(B_R(x_0))} \right\}$$

Then the result naturally follows by a covering argument.  $\square$

**Lemma 6.16.** *Assume  $\phi(t)$  is bounded and non-negative in  $[T_0, T_1]$  with  $T_1 \geq T_0 > 0$ , and for any  $T_0 \leq s < t \leq T_1$ , there holds*

$$\phi(s) \leq \theta\phi(t) + \frac{A}{(t - s)^\alpha} + B$$

where  $\theta, A, B, \alpha$  are nonnegative with  $\theta < 1$ , then

$$\phi(s) \leq C \left\{ \frac{A}{(t - s)^\alpha} + B \right\}$$

The proof is elementary hence omitted.

### 6.3.2 The global estimate

**Lemma 6.17.** *Assume  $u \in W^{2,p}(B_R^+) \cap W_0^{1,p}(B_R^+)$ , and*

$$-\Delta u = f$$

in  $B_R^+$ , then

$$\|D^2u\|_{L^p(B_R^+)} \leq C\|f\|_{L^p(B_R^+)}$$

*Proof.* Through an odd extension, we can define a new function  $\tilde{u}(x)$  in  $W^{2,p}(B_R) \cap W_0^{1,p}(B_R)$ , which satisfies

$$-\Delta \tilde{u} = f$$

in  $B_R$ . Then we only need to apply the result of interior estimate.  $\square$

**Lemma 6.18.** Assume (1)(2)(3), and  $\Omega \subset \mathbb{R}_+^n$ ,  $S \subset \partial\Omega$  and  $S \subset \partial B_R^+$  and  $S$  is flat. If  $u|_S = 0$  and satisfies the linear elliptic equation. Then for  $\Omega' \subset\subset \Omega \cup S$ ,

$$\|D^2u\|_{L^p(\Omega')} \leq \tilde{C} \left\{ \frac{1}{\lambda} \|f\|_{L^p(\Omega)} + \|u\|_{L^p(\Omega')} \right\}$$

where  $\tilde{C} = \tilde{C}(n, p, \lambda, \Lambda, \text{dist}(\Omega', \partial\Omega - S), \tau(a^{ij}))$ .

The proof relies on a covering argument.

**Theorem 6.19.** Assume (1)(2)(3), and  $\partial\Omega$  is  $C^{1,1}$ .  $u \in W^{2,p}(\Omega) \cap W_0^{1,p}(\Omega)$  solves the linear elliptic equation. Then

$$\|D^2u\|_{L^p(\Omega)} \leq \tilde{C} \left\{ \frac{1}{\lambda} \|f\|_{L^p(\Omega)} + \|u\|_{L^p(\Omega)} \right\}$$

where  $\tilde{C} = \tilde{C}(n, p, \lambda, \Lambda, \Omega, \tau(a^{ij}))$ .

The proof is classical, mainly divided into three steps:

1. Interior estimate;
2. Straighten the boundary locally;
3. Covering argument.

*Remark 6.20.* For non-homogeneous boundary condition, i.e.  $u|_{\partial\Omega} = \phi \in W^{2,p}(\Omega)$ , we have  $v = u - \phi$  satisfies the homogeneous boundary condition.

**All the estimates above are a priori estimates. You may wonder how these estimates work. In reality, with continuity method, we can get the existence of solutions to the linear elliptic equation.**

**Idea:**

1. Smooth approximation to the coefficients. Define a sequence of operators  $L_k$  with smooth coefficients which converges to  $L$ .
2. For smooth operators  $L_k$ , we can easily prove the solution  $u_k \in W^{2,p}$ .
3. Apply a priori estimate.
4. Using compactness.

## 7 De Giorgi–Nash–Moser Estimates

### 7.1 Local boundedness

In past lectures, we've introduced some estimates, among which the classical method is to apply 'Freeze the coefficients'. However, it requires the continuity of the principal terms, which cannot be applied to nonlinear elliptic equations directly.

1. In 1957, De Giorgi consider the elliptic equation of divergence form with the principal coefficients  $a^{ij}$  in  $L^\infty$ . His estimates give the Hölder continuity of solutions.
2. In 1958, J.Nash consider the parabolic equation with the principal coefficients  $a^{ij}$  in  $L^\infty$ , which gives similar results.
3. In 1960, J.Moser gives another proof, which can be applied to both elliptic and parabolic equations.

Moser's approach first gives the local boundedness ( $L^\infty$  estimates) of the weak solution, then follows the Hölder continuity of the solution using the weak Harnack inequality.

The estimates above are an important tool in elliptic PDE even now, which is called 'De Giorgi-Nash-Moser Estimates (Iteration)'.

Consider the elliptic equation of the divergence form:

$$Lu = -\operatorname{div}(a^{ij}(x)D_i u) + c(x)u = f(x)$$

in  $B_1$  (We consider the estimates locally).

Assume that

1.  $a^{ij} \in L^\infty(B_1)$ ,  $c(x), f(x) \in L^q(B_1)$  with  $q > \frac{n}{2}$ .

2. For some  $\Lambda > 0$ , we have

$$\|a^{ij}\|_{L^\infty} + \|c\|_{L^q} \leq \Lambda.$$

3.  $(a^{ij})$  is strictly elliptic, i.e.

$$a^{ij}(x)\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall \xi \in \mathbb{R}^n.$$

Let  $u \in H^1(B_1)$  be a subsolution to the equation, i.e.,

$$\int_{B_1} (a^{ij}(x)D_i u D_j \phi + c(x)u\phi) \leq \int_{B_1} f\phi$$

for all  $\phi \in H_0^1(B_1)$ ,  $\phi \geq 0$ .

**Theorem 7.1.** *Assume the hypothesis above,  $u \in H^1(B_1)$  is the weak subsolution. Then  $u^+ \in L_{loc}^\infty(B_1)$ . Moreover, for any  $\theta \in (0, 1)$ ,  $p > 0$ , there exists  $c = c(n, \lambda, \Lambda, p, q) > 0$  such that*

$$\sup_{B_\theta} u^+ \leq c \left\{ \frac{1}{(1-\theta)^{n/p}} \|u^+\|_{L^p(B_1)} + \|f\|_{L^q(B_1)} \right\}$$

### 7.1.1 De giorgi's approach

We first introduce the De giorgi's approach.

*Proof.* 1. First we consider the case that  $\theta = \frac{1}{2}, p = 2$ .

(a) **A proper choice of test function.**

We need to choose a proper test function.

For  $k > 0$ , let  $v = (u - k)^+$ . Then

$$Dv = \begin{cases} Du & u > k \\ 0 & u \leq k \end{cases}$$

Let  $\zeta$  be a cut-off function in  $B_1$  and  $\phi = \zeta^2 v \in H_0^1(B_1)$ ,  $\phi \geq 0$ , then

$$\int_{B_1} (a^{ij}(x) D_i u D_j (\zeta^2 v) + c(x) u \zeta^2 v) \leq \int_{B_1} f \zeta^2 v$$

hence

$$\begin{aligned} \int_{B_1} a^{ij}(x) D_i u D_j (\zeta^2 v) &= \int_{B_1} (a^{ij}(x) D_i u D_j v \zeta^2 + 2a^{ij}(x) D_i u v \zeta D_j \zeta) \\ &= \int_{B_1} (a^{ij}(x) D_i v D_j v \zeta^2 + 2a^{ij}(x) D_i v v \zeta D_j \zeta) \\ &\geq \lambda \int_{B_1} |Dv|^2 \zeta^2 - 2\Lambda \int_{B_1} |Dv| |v| |D\zeta| |\zeta| \\ &\geq \frac{\lambda}{2} \int_{B_1} |Dv|^2 \zeta^2 - c(\Lambda, \lambda) \int_{B_1} |D\zeta|^2 |v|^2 \end{aligned}$$

We use the Yonug inequality in the last step. Note that

$$\begin{aligned} \int_{B_1} c(x) u v \zeta^2 &= \int_{B_1} c(x) (u - k) v \zeta^2 + \int_{B_1} c(x) k v \zeta^2 \\ &\leq \int_{B_1} |c(x)| v^2 \zeta^2 + \int_{B_1} |c(x)| k v \zeta^2 \\ &\leq c \int_{B_1} |c(x)| v^2 \zeta^2 + \int_{B_1} |c(x)| k^2 \zeta^2 \end{aligned}$$

Combining the estimates above with the equation, we get

$$\begin{aligned} \int_{B_1} |Dv|^2 \phi^2 &\leq c(\lambda, \Lambda) \left( \int_{B_1} |v|^2 |D\zeta|^2 + \int_{B_1} |c(x)| v^2 \zeta^2 \right. \\ &\quad \left. + \int_{B_1} |c(x)| k^2 \zeta^2 + \int_{B_1} |f| v \zeta^2 \right) \end{aligned}$$

which implies that

$$\begin{aligned} \int_{B_1} |D(v\zeta)|^2 &\leq c(\lambda, \Lambda) \left( \int_{B_1} v^2 |D\zeta|^2 + \int_{B_1} |c(x)| v^2 \zeta^2 \right. \\ &\quad \left. + \int_{B_1} |c(x)| k^2 \zeta^2 + \int_{B_1} v |f| \zeta^2 \right) \end{aligned}$$

Using Hölder inequality,

$$\int_{B_1} |f| v \zeta^2 \leq \left( \int_{B_1} (v\zeta)^{2^*} \right)^{1/2^*} \left( \int_{v\zeta \neq 0} |f|^{2^{*'}} \right)^{1/2^{*}'}$$

where  $2^* = 2n/(n-2)$ ,  $n \geq 3$ ,  $2^* \in (2, \infty)$ ,  $n = 2$ . By Sobolev inequality ( $v\zeta \in H_0^1(B_1)$ ),

$$\begin{aligned} \int_{B_1} |f|v\zeta^2 &\leq c(n) \left( \int_{B_1} |D(v\zeta)|^2 \right)^{1/2} \left( \int_{\{v\zeta \neq 0\}} |f|^{2^{*'}} \right)^{1/2^{*'}} \\ &\leq \delta \int_{B_1} |D(v\zeta)|^2 + c(\delta, n) \left( \int_{B_1} |f|^{2^{*'}} \right)^{1/2^{*'}} \\ &\leq \delta \int_{B_1} |D(v\zeta)|^2 + c\|f\|_{L^q}^2 |\{v\zeta \neq 0\}|^{1+\frac{2}{n}-\frac{2}{q}} \end{aligned}$$

Since  $1 + \frac{2}{n} - \frac{2}{q} > 1 - \frac{1}{q}$  given that  $q > \frac{n}{2}$ , we get

$$\begin{aligned} \int_{B_1} |D(v\zeta)|^2 &\leq c \left\{ \int_{B_1} v^2 |D\zeta|^2 + \int_{B_1} |c|v^2\zeta^2 + k^2 \int_{B_1} |c|\zeta^2 \right. \\ &\quad \left. + \|f\|_{L^q}^2 |\{v\zeta \neq 0\}|^{1-\frac{1}{q}} \right\} \end{aligned}$$

(b) **An important inequality.** To show that if  $|\{v\zeta \neq 0\}|$  is small enough, there holds

$$\int_{B_1} |D(v\zeta)|^2 \leq c \left\{ \int_{B_1} v^2 |D\zeta|^2 + (k^2 + F^2) |\{v\zeta \neq 0\}|^{1-\frac{1}{q}} \right\} \quad (9)$$

where  $F = \|f\|_{L^q}$ . In fact, when  $c = 0$  the inequality above naturally holds.

In the general case, by Young inequality, Hölder inequality and Sobolev inequality, we get

$$\begin{aligned} \int_{B_1} |c|v^2\zeta^2 &\leq \left( \int_{B_1} |c|^q \right)^{\frac{1}{q}} \left( \int_{B_1} (v\zeta)^{2q'} \right)^{\frac{1}{q'}} \\ &\leq \|c\|_{L^q} \left( \int_{B_1} |v\zeta|^{2^*} \right)^{\frac{2}{2^*}} |\{v\zeta \neq 0\}|^{1-\frac{2}{2^*}-\frac{1}{q}} \\ &\leq c(n) \|c\|_{L^q} \int_{B_1} |D(v\zeta)|^2 |\{v\zeta \neq 0\}|^{\frac{2}{n}-\frac{1}{q}} \end{aligned}$$

and

$$\int_{B_1} |c|\zeta^2 \leq \left( \int_{B_1} |c|^q \right)^{\frac{1}{q}} |\{\zeta \neq 0\}|^{1-\frac{1}{q}} \leq \|c\|_{L^q} |\{v\zeta \neq 0\}|^{1-\frac{1}{q}}$$

Combining the estimates above, we get

$$\begin{aligned} \int_{B_1} |D(v\zeta)|^2 &\leq c \left\{ \int_{B_1} v^2 |D\zeta|^2 + (k^2 + F^2) |\{v\zeta \neq 0\}|^{1-\frac{1}{q}} \right. \\ &\quad \left. + \int_{B_1} |D(v\zeta)|^2 |\{v\zeta \neq 0\}|^{\frac{2}{n}-\frac{1}{q}} \right\} \end{aligned}$$

Note that it implies equation (9) if  $|\{v\zeta \neq 0\}|$  is small enough.

(c) **The iteration inequality.**

By Sobolev inequality and Hölder inequality, we have

$$\int_{B_1} (v\zeta)^2 \leq \left( \int_{B_1} (v\zeta)^{2^*} \right)^{1/2^*} |\{v\zeta \neq 0\}|^{\frac{1}{2^*}} \leq c(n) \int_{B_1} |D(v\zeta)|^2 |\{v\zeta \neq 0\}|^{\frac{2}{n}}$$

Then through the proceeding argument, we get

$$\int_{B_1} (v\zeta)^2 \leq c \left\{ \int_{B_1} v^2 |D\zeta|^2 |\{v\zeta \neq 0\}|^{\frac{2}{n}} + (k^2 + F^2) |\{v\zeta \neq 0\}|^{1+\frac{2}{n}-\frac{1}{q}} \right\}$$

if  $|\{v\zeta \neq 0\}|$  is small enough.

Let  $\varepsilon = \frac{2}{n} - \frac{1}{q}$ , then we get

$$\int_{B_1} (v\zeta)^2 \leq c \left\{ \int_{B_1} v^2 |D\zeta|^2 |\{v\zeta \neq 0\}|^\varepsilon + (k^2 + F^2) |\{v\zeta \neq 0\}|^{1+\varepsilon} \right\} \quad (10)$$

if  $|\{v\zeta \neq 0\}|$  is small enough.

For  $\frac{1}{2} \leq r < R \leq 1$ , we select the cut-off function to satisfy  $\zeta \equiv 1$  in  $B_r$ ,  $\text{supp}\zeta \subset B_R$  and  $|D\zeta| \leq \frac{c}{(R-r)}$ .

Define

$$A(k, r) = \{x \in B_r : u \geq k\}.$$

Note that

$$A(k, R) \leq \frac{1}{k} \int_{A(k, r)} u^+ \leq \frac{c}{k} \|u^+\|_{L^2(B_1)}$$

which can be sufficiently small for  $k$  large enough. For example, we can let  $k \geq k_0 = c\|u^+\|_{L^2(B_1)}$ .

Then (10) holds, and moreover,

$$\int_{A(k, r)} (u - k)^2 \leq c \left\{ \frac{1}{(R - r)^2} \int_{A(k, R)} (u - k)^2 + (k^2 + F^2) |A(k, R)| \right\} |A(k, R)|^\varepsilon$$

For  $h > k \geq k_0, 0 < r < 1$ , there holds  $A(h, r) \subset A(k, r)$  and

$$\int_{A(h, r)} (u - h)^2 \leq \int_{A(k, r)} (u - k)^2,$$

since  $A(h, r) = \{x \in B_r : u \geq h\} = \{x \in B_r : u - k \geq h - k\}$ , we have

$$|A(h, r)| \leq \frac{1}{(h - k)^2} \|u - k\|_{L^2(A(k, r))}^2$$

Therefore,

$$\int_{A(h, r)} (u - h)^2 \leq c \left\{ \frac{1}{(R - r)^2} + \frac{h^2 + F^2}{(h - k)^2} \right\} \frac{1}{(h - k)^{2\varepsilon}} \|u - k\|_{L^2(A(k, r))}^{2+2\varepsilon}$$

Denote  $\phi(h, r) = \|u - h\|_{L^2(A(h, r))}$ , then

$$\phi(h, r) \leq c \left\{ \frac{1}{R - r} + \frac{h + F}{h - k} \right\} \frac{1}{(h - k)^\varepsilon} \phi(k, R)^{1+\varepsilon}$$

(d) **The process of iteration.**

Choose  $\tau = \frac{1}{2}, k > 0$  to be fixed, for  $l = 0, 1, \dots$ , define

$$k_l = k_0 + k \left(1 - \frac{1}{2^l}\right), \quad r_l = 1 + (\tau - 1) \left(1 - \frac{1}{2^l}\right)$$

Then

$$\phi(k_l, r_l) \leq c \left\{ \frac{2^l}{1 - \tau} + \frac{2^l(k_0 + k + F)}{k} \right\} \frac{2^{l\varepsilon}}{k^\varepsilon} (\phi(k_{l-1}, r_{l-1}))^{1+\varepsilon}$$

We claim that for  $k$  large enough, there exists  $t > 1$  such that

$$\phi(k_l, r_l) \leq \frac{\phi(k_0, 1)}{t^l},$$

which can be easily proved through induction. (choose  $t = 2^{1+\frac{1}{\varepsilon}}$ )

Let  $l \rightarrow \infty$ , then  $\phi(k_l, r_l) \rightarrow 0$ , hence  $\phi(k+k_0, 1/2) = 0$ , which implies that  $u^+ \leq k+k_0$  a.e. in  $B_{1/2}$ .

2. We then consider the case that  $\theta \in (0, 1), p \geq 2$ .

In fact, for any  $R < 1$ , we define  $\tilde{u}(y) = u(Ry)$  for  $y \in B_1$ , then

$$\int_{B_1} \tilde{a}^{ij} D_i \tilde{u} D_j \phi + \tilde{c} \tilde{u} \phi \leq \int_{B_1} \tilde{f} \phi$$

for all  $\phi \in H_0^1(B_1), \phi \geq 0$ , where

$$\tilde{a}^{ij}(y) = a^{ij}(Ry), \tilde{c}(y) = R^2 c(Ry), \tilde{f}(y) = R^2 f(Ry).$$

Note that

$$\|\tilde{a}^{ij}\|_{L^\infty(B_1)} + \|\tilde{c}\|_{L^q(B_1)} \leq \|a^{ij}\|_{L^\infty(B_1)} + R^{2-\frac{n}{q}} \|c\|_{L^q(B_1)} \leq \Lambda.$$

Apply the previous case to  $\tilde{u}$ , for any  $p \geq 2$ , we get that

$$\sup_{B_{R/2}} u^+ \leq c \left\{ R^{-\frac{n}{p}} \|u^+\|_{L^p(B_R)} + R^{2-\frac{n}{q}} \|f\|_{L^q(B_R)} \right\}, \quad (11)$$

where we use the Hölder inequality to lift the exponent 2 to  $p$ , and  $c = c(n, \lambda, \Lambda, p, q)$ .

For  $0 < \theta < 1$ , apply (11) on  $B_{(1-\theta)R}(y)$ , to obtain

$$u^+(y) \leq c \left\{ \frac{1}{((1-\theta)R)^{\frac{n}{p}}} \|u^+\|_{L^p(B_{(1-\theta)R}(y))} + ((1-\theta)R)^{2-\frac{n}{q}} \|f\|_{L^q(B_{(1-\theta)R}(y))} \right\},$$

then we get the estimate when  $0 < \theta < 1$  and  $p \geq 2$ .

3. Now we consider the case  $p \in (0, 2)$ .

According to the estimate when  $p = 2$ ,

$$\begin{aligned} \sup_{B_{\theta R}} u^+ &\leq c \left\{ \frac{1}{((1-\theta)R)^{\frac{n}{2}}} \|u^+\|_{L^2(B_R) + ((1-\theta)R)^{2-\frac{n}{q}} \|f\|_{L^q(B_R)}} \right\} \\ &\leq c \left\{ \frac{1}{((1-\theta)R)^{\frac{n}{2}}} \left( \sup_{B_R} u^+ \right)^{\frac{2-p}{2}} \left( \int_{B_R} (u^+)^p \right)^{\frac{1}{2}} + ((1-\theta)R)^{2-\frac{n}{q}} \|f\|_{L^q(B_R)} \right\} \\ &\leq \frac{1}{2} \|u^+\|_{L^\infty(B_R)} + c \left\{ \frac{1}{((1-\theta)R)^{\frac{n}{p}}} \|u^+\|_{L^p(B_R)} + ((1-\theta)R)^{2-\frac{n}{q}} \|f\|_{L^q(B_R)} \right\} \end{aligned}$$

Now we shall use an iteration argument.

Let  $f(t) = \|u^+\|_{L^\infty(B_t)}$ , then  $f(t)$  is non-decreasing and for  $0 < r < R < 1$ , there holds

$$f(r) \leq \frac{1}{2} f(R) + \frac{c}{(R-r)^{\frac{n}{p}}} \|u^+\|_{L^p(B_R)} + c \|f\|_{L^q(B_R)}$$

We can use the iteration lemma in last lecture to acquire,

$$\sup_{B_r} u^+ \leq c \left\{ \frac{1}{(R-r)^{\frac{n}{p}}} \|u^+\|_{L^p(B_R)} + c \|f\|_{L^q(B_R)} \right\}$$

Let  $r = \theta R, R \rightarrow 1$  and we get the estimate when  $p \in (0, 2)$ .

□

### 7.1.2 Moser's approach

Now we introduce the Moser's approach.

*Proof.* 1. First we still consider the case that  $p = 2, \theta = \frac{1}{2}$ . Moser's idea is to establish a reverse Hölder inequality, i.e. for  $p_1 > p_2$  and  $r_1 < r_2$ ,

$$\|u\|_{L^{p_1}(B_{r_1})} \leq c\|u\|_{L^{p_2}(B_{r_2})}$$

(a) **The choice of test function.**

For  $k > 0, m > 0$ , choose  $\bar{u} = u^+ + k$ , and

$$\bar{u}_m = \begin{cases} \bar{u} & u < m \\ k + m & u \geq m \end{cases}$$

then

$$k \leq \bar{u}_m \leq k + m.$$

Then  $D\bar{u}_m = 0$  in  $\{u < 0\} \cup \{u > m\}$ , and  $\bar{u}_m \leq \bar{u}$ .

For  $\beta > 0$  and  $\eta \in C_0(B_1), \eta \geq 0$ , choose the test function

$$\phi = \eta^2(\bar{u}_m^\beta \bar{u} - k^{\beta+1}) \in H_0^1(B_1), \phi \geq 0.$$

We compute  $D\phi$ :

$$D\phi = 2\eta D\eta(\bar{u}_m^\beta \bar{u} - k^{\beta+1}) + \eta^2(\beta \bar{u}_m^{\beta-1} D\bar{u}_m \bar{u} + \bar{u}_m^\beta D\bar{u})$$

Note that  $\phi = 0, D\phi = 0$  in  $\{u < 0\}$ .

For  $k > 0, u^+ \leq \bar{u}, \bar{u}_m^\beta \bar{u} - k^{\beta+1} \leq \bar{u}_m^\beta \bar{u}$ . Then since  $u$  is a subsolution, by ellipticity and Hölder inequality,

$$\begin{aligned} \int_{B_1} a^{ij} D_i u D_j \phi &= \int_{B_1} a^{ij} D_i \bar{u} (\beta D_j \bar{u}_m + D_j \bar{u}) \eta^2 \bar{u}_m^\beta \\ &\quad + 2 \int_{B_1} a^{ij} D_i \bar{u} D_j \eta (\bar{u}_m^\beta \bar{u} - k^{\beta+1}) \eta \\ &\geq \lambda \beta \int_{B_1} \eta^2 \bar{u}_m^\beta |D\bar{u}_m|^2 + \lambda \int_{B_1} \eta^2 \bar{u}_m^\beta |D\bar{u}|^2 \\ &\quad - \Lambda \int_{B_1} |D\bar{u}| |D\eta| \bar{u}_m^\beta \bar{u} \eta \\ &\geq \lambda \beta \int_{B_1} \eta^2 \bar{u}_m^\beta |D\bar{u}_m|^2 + \frac{\lambda}{2} \int_{B_1} \eta^2 \bar{u}_m^\beta |D\bar{u}|^2 \\ &\quad - \frac{2\Lambda^2}{\lambda} \int_{B_1} |D\eta|^2 \bar{u}_m^\beta \bar{u}^2 \end{aligned}$$

where we use Young inequality in the last step. Combining with the lower-order terms and noting  $\bar{u} \geq k$ , we obtain

$$\begin{aligned} \beta \int_{B_1} \eta^2 \bar{u}_m^\beta |D\bar{u}_m|^2 + \int_{B_1} \eta^2 \bar{u}_m^\beta |D\bar{u}|^2 &\leq C \left\{ \int_{B_1} |D\eta|^2 \bar{u}_m^\beta \bar{u}^2 \right. \\ &\quad \left. + \int_{B_1} (|c| \eta^2 \bar{u}_m^\beta \bar{u}^2 + |f| \eta^2 \bar{u}_m^\beta \bar{u}) \right\} \\ &\leq C \left\{ \int_{B_1} |D\eta|^2 \bar{u}_m^\beta \bar{u}^2 + \int_{B_1} c_0 \eta^2 \bar{u}_m^\beta \bar{u}^2 \right\}, \end{aligned}$$

where  $c_0 = |c| + \frac{|f|}{k}$ .

Choose  $k = \|f\|_{L^q}$  if  $f$  is not identically 0. Otherwise choose arbitrary  $k > 0$  and eventually let  $k \rightarrow 0^+$ . By assumption we have

$$\|c_0\|_{L^q} \leq \Lambda + 1.$$

Set  $w = \bar{u}_m^{\frac{\beta}{2}} \bar{u}$  then

$$\begin{aligned} |Dw|^2 &= \left( \frac{\beta}{2} \bar{u}_m^{\frac{\beta}{2}-1} D\bar{u}_m \bar{u} + \bar{u}_m^{\frac{\beta}{2}} D\bar{u} \right)^2 \\ &\leq \frac{\beta^2}{4} |D\bar{u}_m|^2 + \beta \bar{u}_m^{\beta} |D\bar{u}_m|^2 + \bar{u}_m^{\beta} |D\bar{u}|^2 \\ &\leq (1 + \beta) (\beta \bar{u}_m^{\beta} |D\bar{u}_m|^2 + \bar{u}_m^{\beta} |D\bar{u}|^2) \end{aligned}$$

Combining the two estimates above, we get

$$\int_{B_1} |Dw|^2 \eta^2 \leq c(1 + \beta) \left\{ \int_{B_1} w^2 |D\eta|^2 + \int_{B_1} c_0 w^2 \eta^2 \right\}$$

i.e.

$$\int_{B_1} |D(w\eta)|^2 \leq c(1 + \beta) \left\{ \int_{B_1} w^2 |D\eta|^2 + \int_{B_1} c_0 w^2 \eta^2 \right\}$$

(b) **An initial iteration** Through Hölder inequality and Sobolev inequality, we get

$$\begin{aligned} \int_{B_1} |D(w\eta)|^2 &\leq c \left\{ (1 + \beta) \int_{B_1} w^2 |D\eta|^2 + (1 + \beta)^{\frac{2q}{2q-n}} \int_{B_1} w^2 \eta^2 \right\} \\ &\leq c(1 + \beta)^\alpha \int_{B_1} (|D\eta|^2 + \eta^2) w^2 \end{aligned}$$

Then we use the Sobolev inequality,

$$\left( \int_{B_1} (w\eta)^2 \chi \right)^{\frac{1}{\chi}} \leq c(1 + \beta)^\alpha \int_{B_1} (|D\eta|^2 + \eta^2) w^2$$

where  $\chi = \frac{n}{n-2}$ .

Now we choose the cut-off function  $\eta$  such that  $\text{supp} \eta \subset B_R$  and  $\eta \equiv 1$  in  $B_r$  and

$$|D\eta| \leq \frac{2}{R-r}.$$

Then we get the initial iteration inequality:

$$\left( \int_{B_r} w^{2\chi} \right)^{\frac{1}{\chi}} \leq c \frac{(1 + \beta)^\alpha}{(R-r)^2} \int_{B_R} w^2$$

Recalling that  $w = \bar{u}_m^{\frac{\beta}{2}} \bar{u}$ , we get

$$\left( \int_{B_r} \bar{u}_m^{\beta\chi} \bar{u}^{2\chi} \right)^{\frac{1}{\beta\chi}} \leq c \frac{(1 + \beta)^\alpha}{(R-r)^2} \int_{B_R} \bar{u}_m^{\beta} \bar{u}^2$$

or if we let  $\gamma = \beta + 2$

$$\left( \int_{B_r} \bar{u}_m^{\gamma\chi} \right)^{\frac{1}{\chi}} \leq c \frac{(\gamma - 1)^\alpha}{(R-r)^2} \int_{B_R} \bar{u}^\gamma$$

Since  $c$  is independent of  $m$ , we can let  $m \rightarrow \infty$  to get

$$\left( \int_{B_r} \bar{u}^{\gamma\chi} \right)^{\frac{1}{\chi}} \leq c \frac{(\gamma-1)^\alpha}{(R-r)^2} \int_{B_R} \bar{u}^\gamma$$

i.e.

$$\|\bar{u}\|_{L^{\gamma\chi}(B_r)} \leq c \left( \frac{(\gamma-1)^\alpha}{(R-r)^2} \right)^{\frac{1}{\gamma}} \|\bar{u}\|_{L^\gamma(B_R)}$$

- (c) **The process of Iteration** Let  $\gamma_i = 2\chi^i$  and  $r_i = \frac{1}{2} + \frac{1}{2^{i+1}}$ .  
Through an elementary iteration, we have

$$\|\bar{u}\|_{L^{\gamma_i}(B_{r_i})} \leq c \|\bar{u}\|_{L^2(B_1)}$$

Let  $i \rightarrow \infty$  we prove the desired estimate.

2. The general case is the same as the De Giorgi's approach. □

## 7.2 Higher Integrability

We give another application of Moser's iteration.

**Theorem 7.2.** Assume  $n \geq 3$ ,  $a^{ij} \in L^\infty(B_1)$ ,  $c \in L^{n/2}(B_1)$  and

$$\lambda|\xi|^2 \leq a^{ij}(x)\xi_i\xi_j \leq \Lambda|\xi|^2.$$

Let  $u \in H^1(B_1)$  be a subsolution and let  $f \in L^q(B_1)$ , where

$$\frac{2n}{n+2} \leq q < \frac{n}{2}, \quad \frac{1}{q^*} = \frac{1}{q} - \frac{2}{n}.$$

Then  $u^+ \in L_{loc}^{q^*}(B_1)$  and

$$\|u^+\|_{L^{q^*}(B_{1/2})} \leq C \left( \|u^+\|_{L^2(B_1)} + \|f\|_{L^q(B_1)} \right).$$

*Proof.* Set  $\chi = n/(n-2)$ . For  $m > 0$  define

$$\bar{u} = u^+, \quad \bar{u}_m = \min\{\bar{u}, m\}.$$

Let  $\beta \geq 0$  and choose

$$\phi = \eta^2 \bar{u}_m^\beta \bar{u}, \quad 0 \leq \eta \in C_0^1(B_1),$$

as the test function. As in the local boundedness proof, first use a positive regularization of  $\bar{u}$  and then let the regularization tend to zero. By ellipticity and Young's inequality,

$$\begin{aligned} & \beta \int \eta^2 \bar{u}_m^\beta |D\bar{u}_m|^2 + \int \eta^2 \bar{u}_m^\beta |D\bar{u}|^2 \\ & \leq C(1+\beta) \left\{ \int |D\eta|^2 \bar{u}_m^\beta \bar{u}^2 + \int |c| \eta^2 \bar{u}_m^\beta \bar{u}^2 + \int |f| \eta^2 \bar{u}_m^\beta \bar{u} \right\}. \end{aligned}$$

Put

$$w = \bar{u}_m^{\beta/2} \bar{u}.$$

Since

$$|Dw|^2 \leq C(1+\beta) \left( \beta \bar{u}_m^\beta |D\bar{u}_m|^2 + \bar{u}_m^\beta |D\bar{u}|^2 \right),$$

the Sobolev inequality gives

$$\left( \int (\eta w)^{2\chi} \right)^{1/\chi} \leq C(1 + \beta)^2 \left\{ \int (|D\eta|^2 + \eta^2) w^2 + \int |c| \eta^2 w^2 + \int |f| \eta^2 \bar{u}_m^\beta \bar{u} \right\}. \quad (12)$$

We next absorb the two terms which do not occur in the homogeneous iteration. For  $K > 0$ , Hölder and Sobolev inequalities imply

$$\int |c| \eta^2 w^2 \leq K \int \eta^2 w^2 + \left( \int_{\{|c| > K\}} |c|^{n/2} \right)^{2/n} \left( \int (\eta w)^{2\chi} \right)^{1/\chi}.$$

The last coefficient tends to zero as  $K \rightarrow \infty$ . Since the exponents used below remain in a bounded interval, we choose  $K$  once so that the last term is absorbed into the left-hand side of (12).

Let  $\gamma = \beta + 2$ . If

$$2 \leq \gamma \leq \frac{q^*}{\chi} = \frac{q(n-2)}{n-2q},$$

then  $(\gamma - 1)q' \leq \gamma\chi$ . Hence Hölder, interpolation and Young's inequalities give, for every small  $\delta > 0$ ,

$$\begin{aligned} \int |f| \eta^2 \bar{u}_m^\beta \bar{u} &\leq \|f\|_{L^q} \|\eta^2 \bar{u}^{\gamma-1}\|_{L^{q'}} \\ &\leq \delta \left( \int (\eta \bar{u}^{\gamma/2})^{2\chi} \right)^{1/\chi} + C(\delta, \gamma) \left( \int_{\{\eta \neq 0\}} \bar{u}^\gamma + \|f\|_{L^q}^\gamma \right). \end{aligned}$$

We first let  $m \rightarrow \infty$  and then choose  $\delta$  small enough to absorb the first term. If  $\eta = 1$  in  $B_r$ ,  $\eta = 0$  outside  $B_R$  and  $|D\eta| \leq 2/(R-r)$ , we obtain

$$\|\bar{u}\|_{L^{\gamma\chi}(B_r)} \leq C^{1/\gamma} \left\{ (R-r)^{-2/\gamma} \|\bar{u}\|_{L^\gamma(B_R)} + \|f\|_{L^q(B_1)} \right\} \quad (13)$$

for every  $2 \leq \gamma \leq q^*/\chi$ .

Choose an integer  $k \geq 1$  such that

$$2\chi^{k-1} \leq \frac{q^*}{\chi} < 2\chi^k.$$

Apply (13) successively with  $\gamma = 2, 2\chi, \dots, 2\chi^{k-1}$  on nested balls between  $B_1$  and  $B_{3/4}$ . This yields

$$\|\bar{u}\|_{L^{2\chi^k}(B_{3/4})} \leq C \left( \|\bar{u}\|_{L^2(B_1)} + \|f\|_{L^q(B_1)} \right).$$

Since  $q^*/\chi < 2\chi^k$ , the same estimate holds with exponent  $q^*/\chi$  on  $B_{3/4}$ . Finally use (13) once more with  $\gamma = q^*/\chi$ ,  $r = 1/2$  and  $R = 3/4$ . The resulting exponent is  $\gamma\chi = q^*$ , which proves the theorem.  $\square$

### 7.3 Hölder continuity for homogeneous equations

Let

$$Lu = -D_i(a^{ij}D_j u) \text{ in } B_1(0) \subset \mathbb{R}^n$$

Assume the ellipticity condition and  $a^{ij} \in L^\infty(B_1)$ ,  $u \in H^1(B_1)$  is a subsolution.

**Lemma 7.3.** *If  $\Phi \in C_{loc}^{0,1}$  is convex, then*

1.  $v = \Phi(u)$  is a subsolution when  $u$  is a subsolution,  $\Phi' \geq 0$ ,  $v = \Phi(u) \in H_{loc}^1(B_1)$ ,

2.  $v = \Phi(u)$  is a subsolution when  $u$  is a supersolution,  $\Phi' \leq 0, v = \Phi(u) \in H_{loc}^1(B_1)$ .

*Proof.* We first assume that  $\Phi \in C^2$ . If  $u$  is a subsolution and  $\Phi' \geq 0$ , then for every non-negative  $\phi \in C_0^1(B_1)$ ,  $\Phi'(u)\phi$  is an admissible test function. Therefore

$$\begin{aligned} \int a^{ij} D_i v D_j \phi &= \int a^{ij} D_i u D_j (\Phi'(u)\phi) - \int a^{ij} D_i u D_j u \Phi''(u)\phi \\ &\leq 0. \end{aligned}$$

Here the first term is non-positive by the subsolution inequality and the second term is non-positive by convexity. This proves the first statement.

If  $u$  is a supersolution and  $\Phi' \leq 0$ , then  $-\Phi'(u)\phi$  is an admissible non-negative test function. Hence

$$\begin{aligned} 0 &\leq \int a^{ij} D_i u D_j (-\Phi'(u)\phi) \\ &= - \int a^{ij} D_i v D_j \phi - \int a^{ij} D_i u D_j u \Phi''(u)\phi, \end{aligned}$$

which again implies  $\int a^{ij} D_i v D_j \phi \leq 0$ . For a locally Lipschitz convex function, mollify  $\Phi$ , apply the result above, and pass to the limit by dominated convergence.  $\square$

*Remark 7.4.*  $(u - k)^+$  is a subsolution, if  $u$  is a subsolution, since  $\Phi(s) = (s - k)^+ \in C_{loc}^{0,1}(\mathbb{R})$ .

Then we derive a version of Poincaré inequality. We know that the classical Poincaré inequality requires the function to vanish on a portion of the boundary, or we need to subtract the average of the function in the domain. However, the version we are going to derive only requires the function to vanish on a portion of the domain, which is more suitable for our application.

**Lemma 7.5** (A version of Poincaré Inequality). *For any  $\varepsilon > 0$ , there exists  $c = c(n, \varepsilon) > 0$ , such that for  $u \in H^1(B_1)$  with*

$$|\{x \in B_1 : u(x) = 0\}| \geq \varepsilon |B_1|$$

*there holds*

$$\int_{B_1} u^2 \leq c \int_{B_1} |Du|^2$$

*Proof.* Suppose not, then there exists a sequence  $u_m \in H^1(B_1)$  with

$$|\{x \in B_1 : u(x) = 0\}| \geq \varepsilon |B_1|, \quad \int_{B_1} u_m^2 = 1, \quad \int_{B_1} |Du_m|^2 \rightarrow 0.$$

Then  $u_m$  is bounded in  $H^1(B_1)$  hence there exists a subsequence (still denoted by  $u_m$ ), which converges to some function  $u \in H^1(B_1)$  weakly.

By the compact embedding of  $H^1(B_1)$  into  $L^2(B_1)$ , we have  $u_m \rightarrow u$  strongly in  $L^2(B_1)$ . Clearly  $u = u_0$  is nonzero constant. Then

$$0 = \lim_{m \rightarrow \infty} \int_{B_1} |u_0 - u_m|^2 \geq \varepsilon u_0^2 |B_1| > 0$$

which is a contradiction.  $\square$

**Theorem 7.6** (A density theorem). *Suppose  $u$  is a positive supersolution to the elliptic equation in  $B_2$ , while*

$$|\{x \in B_1 : u(x) \geq 1\}| \geq \varepsilon |B_1|$$

*Then there exists  $C = C(n, \frac{\lambda}{\Lambda}, \varepsilon) > 0$ , such that*

$$\inf_{B_{1/2}} u \geq C.$$

*Proof.* For  $\delta > 0$ , let  $u_\delta = u + \delta$ . It is enough to prove an estimate independent of  $\delta$  and then let  $\delta \rightarrow 0^+$ . Define

$$v = (\log u_\delta)^- = \max\{-\log u_\delta, 0\}.$$

The function  $s \mapsto (\log s)^-$  is convex and non-increasing on  $(0, \infty)$ . By the preceding lemma,  $v$  is a non-negative subsolution. Moreover, it is bounded by  $(-\log \delta)^+$ , so the local boundedness theorem applies and gives

$$\sup_{B_{1/2}} v \leq C \left( \int_{B_1} v^2 \right)^{1/2}.$$

Since

$$\{u \geq 1\} \cap B_1 \subset \{v = 0\} \cap B_1,$$

the zero set of  $v$  has measure at least  $\varepsilon|B_1|$ . The preceding Poincaré inequality therefore gives

$$\sup_{B_{1/2}} v \leq C \left( \int_{B_1} |Dv|^2 \right)^{1/2}. \quad (14)$$

It remains to estimate the last integral. Choose  $\zeta \in C_0^1(B_2)$  with  $0 \leq \zeta \leq 1$  and  $\zeta = 1$  in  $B_1$ . Since  $u_\delta$  is a positive supersolution, we may use  $\phi = \zeta^2/u_\delta$  as a test function. Thus

$$\begin{aligned} 0 &\leq \int_{B_2} a^{ij} D_i u_\delta D_j \left( \frac{\zeta^2}{u_\delta} \right) \\ &= 2 \int_{B_2} \zeta a^{ij} D_i \log u_\delta D_j \zeta - \int_{B_2} \zeta^2 a^{ij} D_i \log u_\delta D_j \log u_\delta. \end{aligned}$$

By ellipticity and Young's inequality,

$$\lambda \int_{B_2} \zeta^2 |D \log u_\delta|^2 \leq 2\Lambda \int_{B_2} \zeta |D \log u_\delta| |D\zeta| \leq \frac{\lambda}{2} \int_{B_2} \zeta^2 |D \log u_\delta|^2 + C \int_{B_2} |D\zeta|^2.$$

Consequently,

$$\int_{B_1} |D \log u_\delta|^2 \leq C.$$

On  $\{u_\delta < 1\}$  we have  $Dv = -D \log u_\delta$ , while on  $\{u_\delta \geq 1\}$  we have  $Dv = 0$ . Hence  $\int_{B_1} |Dv|^2 \leq C$ . By (14),  $\sup_{B_{1/2}} v \leq C$ , and therefore

$$\inf_{B_{1/2}} u_\delta \geq e^{-C}.$$

Letting  $\delta \rightarrow 0^+$  proves the theorem.  $\square$

The density theorem is the missing bridge between the information of a level set and a pointwise estimate. We now use it twice, according to which half of the oscillation occupies the larger portion of the ball.

**Theorem 7.7** (Oscillation theorem). *Assume  $u \in H^1(B_2)$  is a bounded weak solution of  $Lu = 0$ . There exists  $\vartheta \in (0, 1)$ , depending only on  $n$  and  $\lambda/\Lambda$ , such that*

$$\operatorname{osc}_{B_{1/2}} u \leq \vartheta \operatorname{osc}_{B_2} u.$$

*Proof.* Let

$$M = \operatorname{ess\,sup}_{B_2} u, \quad m = \operatorname{ess\,inf}_{B_2} u, \quad \omega = M - m.$$

There is nothing to prove if  $\omega = 0$ . Otherwise define

$$v = \frac{u - m}{\omega}, \quad 0 \leq v \leq 1 \quad \text{in } B_2.$$

Since both  $v$  and  $1 - v$  are non-negative solutions, they are also supersolutions. At least one of the following two alternatives holds:

$$|\{v \geq 1/2\} \cap B_1| \geq \frac{1}{2}|B_1|, \quad |\{1 - v \geq 1/2\} \cap B_1| \geq \frac{1}{2}|B_1|.$$

In the first case we apply the density theorem to  $2v$  and obtain  $\inf_{B_{1/2}} v \geq c$ . In the second case we apply it to  $2(1 - v)$  and obtain  $\sup_{B_{1/2}} v \leq 1 - c$ . In either case,

$$\operatorname{osc}_{B_{1/2}} v \leq 1 - c.$$

Thus the desired estimate follows with  $\vartheta = 1 - c$ .  $\square$

**Theorem 7.8** (De Giorgi's Hölder estimate). *Assume  $u \in H^1(B_1)$  is a weak solution of  $Lu = 0$ . Then there exist  $\alpha \in (0, 1)$  and  $C > 0$ , depending only on  $n$  and  $\lambda/\Lambda$ , such that*

$$[u]_{C^\alpha(B_{1/2})} \leq C \|u\|_{L^2(B_1)}.$$

*Proof.* By scaling the oscillation theorem, whenever  $B_{2r}(x_0) \subset B_1$  we have

$$\operatorname{osc}_{B_{r/2}(x_0)} u \leq \vartheta \operatorname{osc}_{B_{2r}(x_0)} u.$$

Thus the oscillation contracts by  $\vartheta$  whenever the radius is divided by four. Iterating, we get

$$\operatorname{osc}_{B_{4^{-k}r}(x_0)} u \leq \vartheta^k \operatorname{osc}_{B_r(x_0)} u.$$

Choose  $\alpha > 0$  such that  $4^{-\alpha} = \vartheta$ . Then for  $0 < \rho < r$ ,

$$\operatorname{osc}_{B_\rho(x_0)} u \leq C \left(\frac{\rho}{r}\right)^\alpha \operatorname{osc}_{B_r(x_0)} u.$$

Finally, the local boundedness theorem controls the oscillation on an interior ball by  $\|u\|_{L^2(B_1)}$ . A covering argument gives the stated estimate.  $\square$

## 7.4 Weak Harnack and Harnack inequalities

The preceding proof of Hölder continuity is De Giorgi's method. Moser's method gives a stronger one-sided estimate for non-negative supersolutions. We first state it for

$$Lu = -D_i(a^{ij}D_j u) = f.$$

The lower-order terms can be treated in the same way as in the local boundedness theorem.

**Theorem 7.9** (Weak Harnack inequality). *Let  $q > n/2$  and let  $u \geq 0$  be a weak supersolution of  $Lu = f$  in  $B_{2R}(x_0)$ . If  $n \geq 3$ , fix any  $0 < p < n/(n - 2)$ ; if  $n = 2$ , fix any  $p > 0$ . Then*

$$\left( \int_{B_R(x_0)} u^p \right)^{1/p} \leq C \left\{ \operatorname{ess\,inf}_{B_R(x_0)} u + R^{2-\frac{n}{q}} \|f\|_{L^q(B_{2R}(x_0))} \right\}.$$

*The constant depends only on  $n, \lambda, \Lambda, q$  and  $p$ .*

*Proof.* By translation and scaling, it is enough to consider  $x_0 = 0$  and  $R = 1$ . Put

$$k = \|f\|_{L^q(B_2)}, \quad \bar{u} = u + k.$$

If  $k = 0$ , we first replace  $u$  by  $u + \delta$  and then let  $\delta \rightarrow 0^+$ . Thus we may assume that  $\bar{u} \geq k > 0$ . The proof has three steps.

### 1. An estimate for the inverse.

Let  $v = \bar{u}^{-1}$ . For any non-negative  $\phi \in H_0^1(B_2) \cap L^\infty(B_2)$ , use  $\bar{u}^{-2}\phi$  as the test function in the supersolution inequality. Since  $Dv = -\bar{u}^{-2}D\bar{u}$ , direct calculation gives

$$\int_{B_2} a^{ij} D_i v D_j \phi + \int_{B_2} \tilde{f} v \phi \leq 0, \quad \tilde{f} = \frac{f}{\bar{u}}.$$

Moreover,

$$\|\tilde{f}\|_{L^q(B_2)} \leq \frac{\|f\|_{L^q(B_2)}}{k} \leq 1.$$

Thus  $v$  is a non-negative subsolution of an equation whose lower-order coefficient has bounded  $L^q$  norm. The local boundedness theorem, applied on nested balls, gives for every  $s > 0$

$$\sup_{B_1} \bar{u}^{-1} \leq C_s \left( \int_{B_{3/2}} \bar{u}^{-s} \right)^{1/s}.$$

Equivalently,

$$\inf_{B_1} \bar{u} \geq C_s^{-1} \left( \int_{B_{3/2}} \bar{u}^{-s} \right)^{-1/s}. \quad (15)$$

### 2. The bridge between positive and negative powers.

We prove that a positive and a negative small moment of  $\bar{u}$  control each other. Let  $B_{2r}(y) \subset B_{7/4}$  and choose a cut-off function  $\eta$  such that

$$\eta = 1 \text{ in } B_r(y), \quad \eta = 0 \text{ outside } B_{2r}(y), \quad |D\eta| \leq \frac{C}{r}.$$

Use  $\eta^2/\bar{u}$  as the test function. We obtain

$$\begin{aligned} 0 &\leq \int a^{ij} D_i \bar{u} D_j \left( \frac{\eta^2}{\bar{u}} \right) - \int f \frac{\eta^2}{\bar{u}} \\ &= 2 \int \eta a^{ij} D_i \log \bar{u} D_j \eta - \int \eta^2 a^{ij} D_i \log \bar{u} D_j \log \bar{u} - \int \tilde{f} \eta^2. \end{aligned}$$

By ellipticity, Young's inequality and  $\|\tilde{f}\|_{L^q} \leq 1$ ,

$$\int_{B_r(y)} |D \log \bar{u}|^2 \leq C r^{n-2}.$$

The Poincaré inequality yields

$$\int_{B_r(y)} \left| \log \bar{u} - (\log \bar{u})_{B_r(y)} \right| \leq C.$$

Hence  $\log \bar{u}$  has a universal BMO norm in  $B_{7/4}$ . By the John–Nirenberg inequality there exists  $p_0 > 0$  such that, if

$$w = \log \bar{u} - (\log \bar{u})_{B_{3/2}},$$

then

$$\int_{B_{3/2}} e^{p_0|w|} \leq C.$$

Since  $e^{p_0 w} = e^{-p_0(\log \bar{u})_{B_{3/2}}} \bar{u}^{p_0}$  and  $e^{-p_0 w} = e^{p_0(\log \bar{u})_{B_{3/2}}} \bar{u}^{-p_0}$ , we conclude

$$\left( \int_{B_{3/2}} \bar{u}^{p_0} \right) \left( \int_{B_{3/2}} \bar{u}^{-p_0} \right) \leq C. \quad (16)$$

Combining (15), with  $s = p_0$ , and (16), we get

$$\left( \int_{B_{3/2}} \bar{u}^{p_0} \right)^{1/p_0} \leq C \inf_{B_1} \bar{u}. \quad (17)$$

### 3. Lifting the exponent.

We now show that  $p_0$  can be replaced by every exponent in the theorem. Let  $0 < \gamma < 1$ , put  $\beta = 1 - \gamma$ , and use  $\eta^2 \bar{u}^{-\beta}$  as the test function. After expanding the derivative and using Young's inequality, we obtain

$$\int \eta^2 \bar{u}^{\gamma-2} |D\bar{u}|^2 \leq \frac{C}{(1-\gamma)^2} \int \bar{u}^\gamma (|D\eta|^2 + \eta^2).$$

Here the forcing term is written as  $|f| \bar{u}^{-\beta} = |\tilde{f}| \bar{u}^\gamma$  and is estimated by Hölder, interpolation and Sobolev inequalities. With  $z = \bar{u}^{\gamma/2}$ , the preceding estimate implies

$$\int |D(z\eta)|^2 \leq \frac{C}{(1-\gamma)^a} \int z^2 (|D\eta|^2 + \eta^2)$$

for some  $a = a(n, q) > 0$ . Therefore, with  $\chi = n/(n-2)$  when  $n \geq 3$ , Sobolev inequality gives

$$\|\bar{u}\|_{L^{\gamma\chi}(B_r)} \leq \left( \frac{C}{(1-\gamma)^a (R-r)^2} \right)^{1/\gamma} \|\bar{u}\|_{L^\gamma(B_R)} \quad (18)$$

for  $1 < r < R < 3/2$ . In dimension two we use an arbitrary finite Sobolev exponent instead of  $\chi$ .

When  $n \geq 3$ , fix  $p < \chi$ . Choose an integer  $m$  such that  $s_0 = p\chi^{-m} \leq p_0$  and set  $s_j = \chi^j s_0$ . Then  $s_j < 1$  for  $j < m$ , while  $s_m = p$ . Starting from (17), apply (18) finitely many times on nested balls from  $B_{3/2}$  to  $B_1$ . We obtain

$$\left( \int_{B_1} \bar{u}^p \right)^{1/p} \leq C \inf_{B_1} \bar{u}.$$

When  $n = 2$ , first choose a finite Sobolev exponent  $\chi > p$  and run the same finite iteration. Since  $\bar{u} = u + k$ , this becomes

$$\left( \int_{B_1} u^p \right)^{1/p} \leq C \left( \inf_{B_1} u + \|f\|_{L^q(B_2)} \right).$$

Returning to the original scale replaces the last term by  $R^{2-n/q} \|f\|_{L^q(B_{2R})}$  and finishes the proof. □

*Remark 7.10.* The weak Harnack inequality is an estimate for supersolutions: an  $L^{p_0}$  average is controlled by an infimum. The local boundedness theorem is the opposite one-sided estimate for subsolutions: a supremum is controlled by an  $L^p$  average. Their combination is the Harnack inequality for solutions.

**Theorem 7.11** (Moser's Harnack inequality). *Let  $u \geq 0$  be a weak solution of  $Lu = f$  in  $B_{2R}(x_0)$ . Under the preceding assumptions,*

$$\operatorname{ess\,sup}_{B_R(x_0)} u \leq C \left\{ \operatorname{ess\,inf}_{B_R(x_0)} u + R^{2-\frac{n}{q}} \|f\|_{L^q(B_{2R}(x_0))} \right\}.$$

*Proof.* Apply the local boundedness theorem to the non-negative subsolution  $u$ :

$$\sup_{B_R} u \leq C \left\{ \left( \int_{B_{3R/2}} u^{p_0} \right)^{1/p_0} + R^{2-\frac{n}{q}} \|f\|_{L^q(B_{2R})} \right\}.$$

The weak Harnack inequality controls the first term by the infimum of  $u$  and the same forcing term. This proves the theorem.  $\square$

*Remark 7.12* (Two routes for supersolutions). The proof above is Moser's negative-power method: the BMO estimate for  $\log u$  connects a positive moment to a negative moment. There is also a pure De Giorgi route. In that proof, Caccioppoli and Sobolev inequalities first give an expansion of positivity, and a covering argument then gives the weak Harnack inequality. The conclusion is the same, but the intermediate mechanisms are different.

## 8 Krylov–Safonov and Evans–Krylov Theory

### 8.1 Krylov–Safonov theory

Unlike the De Giorgi–Nash–Moser theory, the Krylov–Safonov theory deals with equations in non-divergence form and derives the corresponding Harnack inequality and Hölder continuity.

We first state the main conclusion of the Krylov–Safonov theory.

**Theorem 8.1** (Krylov–Safonov Harnack inequality). *Let  $u$  be a nonnegative solution of*

$$Lu = \sum_{i,j=1}^n a^{ij}(x) D_{ij}u = 0 \quad \text{in } B_1 \subset \mathbb{R}^n,$$

where

$$\lambda I \leq (a^{ij}(x)) \leq \Lambda I.$$

Then

$$\sup_{B_{1/2}} u \leq C \inf_{B_{1/2}} u,$$

where  $C = C(\lambda, \Lambda, n)$ .

The idea of the proof is similar to that in the De Giorgi–Nash–Moser theory.

#### 8.1.1 Weak Harnack inequality

We first derive a weak Harnack inequality.

**Theorem 8.2** (Weak Harnack inequality). *If  $u$  is a nonnegative supersolution of the equation in  $B_1$ , then there exists  $\varepsilon > 0$  such that*

$$\|u\|_{L^\varepsilon(B_{1/2})} \leq C \inf_{B_{1/2}} u,$$

where  $C = C(\lambda, \Lambda, n)$ .

The idea is to use weak  $L^p$  spaces. By the slice lemma,  $L_w^p \subset L^q$  whenever  $p > q$ . Hence it is enough to prove that there exists  $\varepsilon' > 0$  such that

$$\|u\|_{L_w^{\varepsilon'}(B_{1/2})} \leq C \inf_{B_1} u.$$

After normalizing the infimum of  $u$ , it suffices to prove the following lemma.

**Lemma 8.3.** *Suppose that  $v \geq 0$  is a supersolution in  $B_1$  and  $v(0) = 1$ . Then there exists a universal  $\varepsilon > 0$  such that*

$$|\{v > t\} \cap B_{1/2}| \leq t^{-\varepsilon} \quad \text{for every } t > 0.$$

Equivalently,  $v \in L_w^\varepsilon(B_{1/2})$ .

We also need the following Calderón–Zygmund type lemma to start the iteration. In some contexts it is called the ink-spots lemma.

**Lemma 8.4.** *Let  $A \subset B \subset Q_1$  be measurable sets and let  $0 < \delta < 1$ . Assume that  $|A| \leq \delta$ . Suppose that, whenever a subcube  $Q \subset Q_1$  satisfies*

$$|A \cap Q| > \delta|Q|,$$

*a fixed enlargement  $\tilde{Q}$  of  $Q$  is contained in  $B$ . Then*

$$|A| \leq \delta|B|.$$

*Proof.* Perform the dyadic Calderón–Zygmund decomposition of  $A$  at height  $\delta$ . Let  $\{Q_j\}$  be the selected dyadic cubes, and let  $Q_j^*$  be their dyadic predecessors. By maximality,

$$|A \cap Q_j^*| \leq \delta |Q_j^*|.$$

Moreover,  $Q_j^*$  is contained in the fixed enlargement  $\tilde{Q}_j$ , so the assumption implies that  $Q_j^* \subset B$ . After selecting a disjoint family of maximal predecessors, still denoted by  $\{Q_j^*\}$ , which covers  $A$  up to a null set, we obtain

$$|A| \leq \sum_j |A \cap Q_j^*| \leq \delta \sum_j |Q_j^*| \leq \delta |B|.$$

□

*Remark 8.5.* In De Giorgi’s approach to the weak Harnack inequality, this lemma is also used.

We now recall the ABP estimate proved in the previous note. The ABP estimate plays the role of the Sobolev inequality in the Krylov–Safonov theory, since both convert information from the equation into information about the measure of a level set.

**Theorem 8.6** (ABP estimate). *If  $v$  satisfies*

$$a^{ij} D_{ij} v = f \quad \text{in } B_1, \quad v \geq 0 \quad \text{on } \partial B_1,$$

*then*

$$(\sup_{B_1} v^-)^n \lesssim \int_{\Gamma^+} |f^+|^n,$$

*where  $\Gamma^+$  is the contact set.*

Using the ABP estimate, we derive two contact estimates: one starts the iteration, and the other advances it.

**Lemma 8.7** (Contact lemma 1). *Under the same assumptions on  $v$ , there exists a universal  $\theta > 0$  such that*

$$|\{v < 2\} \cap B_1| \geq \theta.$$

*Proof.* Set

$$\tilde{v} = v + 2(|x|^2 - 1).$$

Then  $\tilde{v}(0) = -1$  and  $\tilde{v} \geq 0$  on  $\partial B_1$ . Moreover, the supersolution inequality gives

$$L\tilde{v} \leq C.$$

Applying the ABP estimate to  $\tilde{v}$ , we obtain

$$1 \leq (\sup_{B_1} \tilde{v}^-)^n \leq C^n |\Gamma^+|.$$

On the contact set  $\Gamma^+$ , one has  $\tilde{v} < 0$ , and hence  $v < 2$ . Therefore

$$1 \leq C^n |\{v < 2\} \cap B_1|,$$

which proves the claim. □

**Lemma 8.8** (Contact lemma 2). *Under the same assumptions on  $v$ , let  $x_0 \in B_{1/4}$  and let  $M > 0$  be a sufficiently large universal constant. Then there exists a universal  $\theta > 0$  such that*

$$|\{v < M\} \cap B_{1/8}(x_0)| \geq \theta |B_{1/8}(x_0)|.$$

*Proof.* Construct a function

$$\gamma(x) = \begin{cases} \frac{\varepsilon}{2}|x - x_0|^2 - A, & x \in B_{1/8}(x_0), \\ -|x - x_0|^{-\sigma} - B, & x \notin B_{1/8}(x_0), \end{cases}$$

where  $A$  and  $B$  are chosen so that the two pieces agree, and  $\sigma$  is chosen sufficiently large. Then

$$L\gamma \leq C \quad \text{in } B_{1/8}(x_0), \quad L\gamma \leq 0 \quad \text{outside } B_{1/8}(x_0).$$

Set  $\tilde{v} = v + \gamma$ . We may arrange that  $\tilde{v}(0) = -2$  and  $\inf \gamma \geq -M$ , where  $M > 2$ . Applying the ABP estimate gives

$$2^n \leq (\sup \tilde{v}^-)^n \leq C^m |\Gamma^+|.$$

The contact set  $\Gamma^+$  cannot lie outside  $B_{1/8}(x_0)$ , since  $\tilde{v}$  is a supersolution there. On  $\Gamma^+$  we also have  $\tilde{v} < 0$  and  $\gamma \geq -M$ , so  $v < M$ . Consequently,

$$2^n \leq C^m |\{v < M\} \cap B_{1/8}(x_0)|,$$

and the result follows. □

*Proof of Lemma 8.3.* Let

$$A_k = \{v > M^k\} \cap B_{1/2}.$$

We apply Lemma 8.4 to the pair  $(A_k, A_{k-1})$  with  $\delta = 1 - \theta$ . Suppose that a cube  $Q$  satisfies

$$\frac{|A_k \cap Q|}{|Q|} > 1 - \theta.$$

Equivalently,

$$|\{v \leq M^k\} \cap Q| < \theta|Q|.$$

After scaling  $v/M^{k-1}$  and rescaling  $Q$  to  $B_{1/8}(x_0)$ , Lemma 8.8 shows that the fixed enlargement  $\tilde{Q}$  must be contained in  $A_{k-1}$ ; otherwise the preceding inequality is contradicted. Thus the hypotheses of Lemma 8.4 hold, and

$$|A_k| \leq (1 - \theta)|A_{k-1}|.$$

Iteration gives

$$|A_k| \leq (1 - \theta)^k |A_0|.$$

Choose  $\varepsilon > 0$  so that  $M^{-\varepsilon} = 1 - \theta$ . Then iteration gives that

$$|\{v > t\} \cap B_{1/2}| \leq Ct^{-\varepsilon},$$

which is the desired weak  $L^\varepsilon$  estimate. □

### 8.1.2 Local boundedness

**Theorem 8.9** (Local boundedness). *Let  $u \geq 0$  be a subsolution in  $B_1$ . For every  $q > 0$ ,*

$$\sup_{B_{1/2}} u \leq C_q \left( \int_{B_1} u^q \right)^{1/q}.$$

*Proof.* After dividing  $u$  by

$$N = \left( \int_{B_1} u^q \right)^{1/q},$$

we may assume that  $N = 1$ . Chebyshev's inequality then gives

$$|\{u > t\} \cap B_1| \leq t^{-q} |B_1| \quad (t > 0).$$

We prove that  $u$  is universally bounded in  $B_{1/2}$ .

Suppose that  $u(x_0) = K$  for some  $x_0 \in B_{1/2}$ . Fix  $0 < \beta < q/n$  and put  $r = K^{-\beta}$ . If  $K$  is sufficiently large, then  $Q_r(x_0) \subset B_{3/4}$ . We claim that there exists a universal  $\gamma > 0$  such that

$$\sup_{Q_r(x_0)} u > (1 + \gamma)K.$$

Otherwise,

$$w = \frac{(1 + \gamma)K - u}{\gamma K}$$

is a nonnegative supersolution in  $Q_r(x_0)$  and  $w(x_0) = 1$ . Rescale  $Q_r(x_0)$  to a unit cube and apply Lemma 8.3. In the concentric half cube, let

$$E_+ = \{u \geq K/2\}, \quad E_- = \{u < K/2\}.$$

The tail estimate for  $u$  gives

$$|E_+| \leq CK^{-q},$$

while  $|Q_{r/2}(x_0)| = c_n K^{-n\beta}$ . Since  $n\beta < q$ , for sufficiently large  $K$ ,

$$|E_+| < \frac{1}{4} |Q_{r/2}(x_0)|.$$

On  $E_-$ ,

$$w > \frac{1/2 + \gamma}{\gamma}.$$

If  $\varepsilon$  is the exponent in Lemma 8.3, then

$$|E_-| \leq C \left( \frac{\gamma}{1/2 + \gamma} \right)^\varepsilon |Q_{r/2}(x_0)|.$$

Choose  $\gamma$  so small that the right-hand side is less than  $\frac{1}{4} |Q_{r/2}(x_0)|$ . This is impossible because  $E_+$  and  $E_-$  partition  $Q_{r/2}(x_0)$ . The claim follows.

Starting from a sufficiently large value  $K_0 = u(x_0)$  and repeating the claim, we find points  $x_j$  and values  $K_j = u(x_j)$  such that

$$K_{j+1} \geq (1 + \gamma)K_j, \quad |x_{j+1} - x_j| \leq CK_j^{-\beta}.$$

But

$$\sum_{j=0}^{\infty} |x_{j+1} - x_j| \leq CK_0^{-\beta} \sum_{j=0}^{\infty} (1 + \gamma)^{-j\beta}.$$

For  $K_0$  large enough, all  $x_j$  remain in a fixed compact subset of  $B_1$ , whereas  $K_j \rightarrow \infty$ , a contradiction. Hence  $\sup_{B_{1/2}} u \leq C_q$  under the normalization  $N = 1$ . Scaling back proves the theorem.  $\square$

Combining the weak Harnack inequality and local boundedness gives the Harnack inequality. Hölder continuity then follows from the Harnack inequality by iteration.

**Corollary 8.10** (Hölder continuity). *There exists  $\alpha > 0$  such that every solution  $v$  satisfies*

$$|v|_{C^\alpha(B_{1/2})} \leq C \|v\|_{L^\infty(B_1)}.$$

## 8.2 Evans–Krylov theory

The Evans–Krylov theory is based on the Krylov–Safonov theory and mainly concerns nonlinear elliptic equations

$$F(D^2u) = 0.$$

Choosing a direction  $e$  and differentiating once, we obtain the linear elliptic equation

$$F_{ij}(D^2u)D_{ij}u_e = 0. \quad (1)$$

The  $C^{1,\alpha}$  estimate for such an equation is established by the Krylov–Safonov theory.

The Evans–Krylov theory gives a much stronger result.

**Theorem 8.11** (Evans–Krylov theorem). *Suppose that  $F$  is uniformly elliptic and either concave or convex,  $F(0) = 0$ , and that  $u$  solves  $F(D^2u) = 0$  in  $B_1$ . Then*

$$\|u\|_{C^{2,\alpha}(B_{1/2})} \leq C\|u\|_{L^\infty(B_1)}.$$

*Proof.* The proof is divided into three steps. We take concavity as an example.

### Step I. A priori estimate.

First we take derivative to the direction  $e$  twice, we get

$$0 = F_{ij,kl}(D^2u)D_{ij}u_eD_{kl}u_e + F_{ij}D_{ij}u_{ee} \leq F_{ij}(D^2u)D_{ij}u_{ee}.$$

Therefore,  $u_{ee}$  is a subsolution to an uniformly elliptic equation.

We choose  $e$  to be the orthonormal basis, and we get that  $\Delta u$  is also a subsolution. Furthermore, without loss of generality, we may assume that  $F_{ij}(0) = \delta_{ij}$ ,  $F(0) = 0$ , otherwise we can compose  $F$  with a affine transformation. Then the concavity of  $F$  gives that

$$F(D^2u) \leq F(0) + F_{ij}(0)(D^2u)_{ij} = \Delta u,$$

which means that  $\Delta u$  is also a nonnegative solution.

Then from the local boundedness in Krylov–Safonov theory, we get that

$$\|\Delta u\|_{L^\infty(B_{1/2})} \leq C\|\Delta u\|_{L^\varepsilon(B_{3/4})},$$

for some  $\varepsilon$ .

We choose  $\Delta u$  because  $\Delta$  has a special divergence structure, i.e. we choose a cutoff function supported on  $B_1$  and  $\zeta \equiv 1$  on  $B_{3/4}$ . Then

$$\|\Delta u\|_{L^1(B_{3/4})} = \int_{B_1} \zeta \Delta u = \int_{B_1} u \Delta \zeta \leq C\|u\|_{L^\infty(B_1)}.$$

Since  $L^1 \subset L^\varepsilon$  in a finite measure set, we deduce that

$$\|\Delta u\|_{L^\infty(B_{1/2})} \leq C\|u\|_{L^\infty(B_1)}.$$

Then we will use the  $L^p$  theory. To be specific, we consider the special Poisson equation

$$\Delta u = f, \quad f = \Delta u \in L_{loc}^\infty,$$

then the Calderon–Zygmund estimates give that

$$\|D^2u\|_{L^p(B_{3/5})} \leq C(\|u\|_{L^p(B_{3/4})} + \|\Delta u\|_{L^p(B_{3/4})}) \leq C\|u\|_{L^\infty(B_1)}.$$

Then we get the  $L^p$  norm of any second-order derivative  $u_{ee}$ . Apply the local boundedness again to  $u_{ee}^+$ , which is a nonnegative subsolution, we get

$$\|u_{ee}^+\|_{L^\infty(B_{1/2})} \leq \|u_{ee}^+\|_{L^\varepsilon(B_{3/5})} \leq C\|u\|_{L^\infty(B_1)}.$$

Then we need to evaluate the negative part. Since we have  $\Delta u = u_{ee} + u_{e_2 e_2} + \cdots + u_{e_n e_n} \geq 0$ , where  $e, e_2, \dots, e_n$  are orthonormal basis, then for  $x \in B_1$  such that  $u_{ee}(x) < 0$ , we have

$$-u_{ee}(x) \leq \sum_{i=2}^n u_{e_i e_i}^+(x) \leq C \|u\|_{L^\infty(B_1)},$$

which gives the estimate of the  $[D^2 u]_\infty$  seminorm. (We need to remark that the argument above is not strict, since we don't know whether  $u_{ee}$  does exist. However, this doesn't matter since we introduce the method of difference quotient in the previous note which is useful here.)

**Step II. A Geometrical Interpretation.** To get the  $C^{2,\alpha}$  estimate, it suffices to prove the oscillation control like

$$\text{osc}_{B_r} D^2 u \leq \rho \text{osc}_{B_1} D^2 u$$

We can normalize this situation that if  $\text{osc}_{B_1} D^2 u = 1$ , then there exists  $r_0$  such that

$$\text{osc}_{B_{r_0}} D^2 u \leq \frac{1}{2}.$$

In a geometric viewpoint, if we regard  $D^2 u$  as a map from  $B_1$  to the space of symmetry matrices, then it is equivalent to prove

$$\text{diam}(D^2 u(B_{r_0})) \leq \frac{1}{2}.$$

given that  $\text{diam}(D^2 u(B_1)) = 1$ .

We denote  $\Gamma$  as the image set  $\{D^2 u(B_1)\}$ . Then  $\text{diam} \Gamma = 1$  implies that there are two matrices  $M_0, M_1$  such that

$$\|M_1 - M_0\| = 1.$$

A useful observation is that, by the uniform ellipticity of  $F$ ,

$$F(M_1) - F(M_0) = F_{ij}(M_1 - M_0)_{ij},$$

or

$$F_{ij}(M_1 - M_0)_{ij}^+ = F_{ij}(M_1 - M_0)_{ij}^-,$$

which implies that,  $\|M_1 - M_0\|, \|(M_1 - M_0)^+\|, \|(M_1 - M_0)^-\|$  are of the same order.

Back to the argument, we do two covering of  $\Gamma$  by balls  $B_j(\delta)$  and  $\hat{B}_j(\varepsilon)$ , with  $\varepsilon \ll \delta$  to be fixed.

We denote the inverse map of  $D^2 u(\cdot)$  as  $\mathcal{H}^{-1}$ . And  $\mathcal{H}^{-1}(B_j)$  covers  $B_1$ , since the number of the preimage balls is of the order  $\delta^{-n^2}$ . Then by the pigeon hole principle, we conclude that there exists  $B^0$  among  $\{B_j\}$  that

$$|\mathcal{H}^{-1}(B^0)| \geq \delta^{n^2}.$$

We use  $\bar{M}$  to denote the element in  $B^0$ . Since  $\text{diam} \mathcal{H}(B_1) = 1 \geq 1/2$ , there exists  $M_2$  such that

$$\|M_2 - \bar{M}\| \geq \frac{1}{4}.$$

(Here we use  $1/2$  instead of  $1$ , because it is helpful to continue the following deletion argument.)

The previous observation gives that

$$\|(M_2 - \bar{M})^+\| \geq \theta, \quad (*)$$

where  $\theta$  is a constant dependent only on the ellipticity of  $F$ .

Assume that  $M_2 = D^2 u(x_2)$ ,  $\bar{M} = D^2 u(\bar{x})$ , then  $(*)$  implies that there exists a direction  $\alpha$  such that

$$D_{\alpha\alpha}^2 u(x_2) \geq D_{\alpha\alpha}^2 u(\bar{x}) + \theta.$$

If we fixed  $\delta$  sufficiently small (i.e.  $\delta \ll \theta$ ), we have

$$D_{\alpha\alpha}^2 u(x_2) \geq D_{\alpha\alpha}^2 u(\bar{x}) + \frac{\theta}{2},$$

where  $x \in \mathcal{H}^{-1}(B^0)$ . Therefore,

$$\sup_{B_1} D_{\alpha\alpha}^2 u(x) \geq \sup_{\mathcal{H}^{-1}(B^0)} \left( D_{\alpha\alpha}^2 u(x) + \frac{\theta}{2} \right).$$

**Step III. The deletion argument.** Notice that

$$w = \sup_{B_1} D_{\alpha\alpha}^2 u(x) - D_{\alpha\alpha}^2 u(x)$$

is a nonnegative supersolution, we can apply the weak Harnack inequality to acquire

$$\inf_{B_{1/2}} w \geq \left( \int_{B_1} w^\varepsilon \right)^{1/\varepsilon} \geq \left( \int_{\mathcal{H}^{-1}(B^0)} \frac{\theta}{2} \right)^{1/\varepsilon} \geq C\theta^{1/\varepsilon},$$

i.e.

$$\sup_{B_{1/2}} D_{\alpha\alpha}^2 u \leq \sup_{B_1} D_{\alpha\alpha}^2 u - C\theta^{1/\varepsilon}.$$

We now, finally, consider the family of balls of radius  $\varepsilon$ . If  $\varepsilon$  is sufficiently small (i.e.  $\varepsilon \ll C\theta^{1/\varepsilon}$ ), there must be some ball that disjoint  $\mathcal{H}(B_{1/2})$ .

Then if  $\text{diam}(D^2 u(B_{1/2})) \leq \frac{1}{2}$ , we get the initial result. If not, we can rescale  $\tilde{u} = 2^2 u(x/2)$ , then  $\tilde{u}$  satisfies the same properties, and all the arguments above can be repeated, which means we can remove another ball which is disjoint with  $\tilde{\mathcal{H}}(B_{1/2}) = \mathcal{H}(B_{1/4})$ .

This process must stop in finite steps since the number of the ball of radius  $\varepsilon$  is finite. i.e. We can always find some  $r_0$  such that  $\text{diam}(B_{r_0}) \leq \frac{1}{2}$ .

**Step IV. The iteration process.** We have got the iteration

$$\text{osc}_{B_r} D^2 u \leq \rho \text{osc}_{B_1} D^2 u.$$

Since  $\text{osc}_{B_1} D^2 u$  is controlled in Step I, we can finally derive the Hölder estimate.

□

## References

- [1] Q. Han and F. Lin, *Elliptic Partial Differential Equations*, 2nd ed., Courant Lecture Notes in Mathematics 1, American Mathematical Society, 2011.
- [2] D. Gilbarg and N. S. Trudinger, *Elliptic Partial Differential Equations of Second Order*, 2nd ed., Springer, 1983.
- [3] L. C. Evans, *Partial Differential Equations*, 2nd ed., Graduate Studies in Mathematics 19, American Mathematical Society, 2010.
- [4] L. A. Caffarelli and X. Cabré, *Fully Nonlinear Elliptic Equations*, American Mathematical Society Colloquium Publications 43, 1995.
- [5] L. Caffarelli, *Lecture Notes*, version of December 12, 2013.
- [6] Z. Qian, *Note on Regularity Theory for Nonlinear Elliptic Equations*, arXiv:2512.03012.
- [7] L. A. Caffarelli, B. Gidas, and J. Spruck, *Asymptotic symmetry and local behavior of semilinear elliptic equations with critical Sobolev growth*, Comm. Pure Appl. Math. 42 (1989), no. 3, 271–297.